



Remote sensing facilitates the exploration of algal bloom dynamics and its hydrometeorological drivers in tributary bays of the Three Gorges Reservoir

Aiping Jiang^a, Dongsheng Wang^b, Tiantian Jin^a, Peng Li^c, Tao Xu^c, Di Zhang^a, Boran Zhu^a, Junqiang Lin^a, Qidong Peng^{a,*}

^a State Key Laboratory of Water Cycle and Water Security, China Institute of Water Resources and Hydropower Research, Beijing, 100038, China

^b China Renewable Energy Engineering Institute, Beijing, 100120, China

^c China Yangtze Power Co., Ltd., Wuhan, 430010, China

ARTICLE INFO

Keywords:

Algal bloom
Three Gorges Reservoir (TGR)
Sentinel-2
Google Earth Engine (GEE)
Xiangxi bay (XXB)
Normalized Difference Chlorophyll Index (NDCI)

ABSTRACT

Reservoir construction has led to frequent algal blooms in tributaries of backwater areas, threatening water quality, aquatic ecological safety, and public health. The mechanisms governing algal blooms in reservoir bays are complex, leading to spatiotemporal occurrence patterns that are difficult to accurately identify. In this study, Sentinel-2 imagery from 2019 to 2023 were used to characterize the spatiotemporal distribution of algal blooms in Xiangxi Bay (XXB) near the dam of the Three Gorges Reservoir (TGR). The Google Earth Engine (GEE) platform was used for remote sensing image selection, preprocessing, and spectral index calculation, while random forest (RF) model and polynomial regression were employed to develop chlorophyll-a (Chl-a) retrieval model. Box plots and the Mann–Whitney *U* test were used to compare bloom and non-bloom conditions across nine hydrometeorological variables from the perspective of external influences, while the internal bloom mechanisms were analyzed by integrating the critical depth theory of algal growth with the hydrodynamic mixing characteristics and bloom occurrence patterns in XXB. The results indicated that: (1) Sentinel-2 imagery effectively captures bloom dynamics in narrow tributary embayments, and the Normalized Difference Chlorophyll Index (NDCI) accurately retrieves Chl-a, with the retrieval model achieving an R^2 of 0.76; (2) In recent years, algal blooms in XXB have been more likely to occur in March, July, August, and September, particularly in August, with approximately 18 % of the XXB experiencing blooms, predominantly at level II intensity. High-frequency bloom areas are mainly distributed in the mid-to upper reaches of the backwater zone; (3) During the flood season, mid-layer intrusion from the Yangtze River and upstream tributary inflow promote stable stratification in XXB, especially under low-rainfall conditions, thereby favoring algal bloom formation in the mid-to upper reaches. Enhancing hydrodynamic disturbance through physical mixing or reservoir inflow and water-level regulation, particularly under low 10-day rainfall (<35 mm), can effectively suppress bloom development.

1. Introduction

Algal blooms typically occur in oceans, lakes, and reservoirs, but are relatively rare in riverine systems (Huo et al., 2022; Xiao et al., 2019). In recent years, however, climate change and reservoir construction have increased the frequency of blooms in reservoir tributaries, posing growing ecological and public health risks. Globally, more than 58,000 dams have been built, and this number continues to rise due to increasing water management demands and clean energy development

(Mulligan et al., 2020; Wang et al., 2024). While large dams provide benefits such as flood control and hydropower, they fundamentally alter hydrological processes and flow regimes. These changes, together with nutrient enrichment, thermal stratification, and increased light availability, create conditions conducive to algal blooms. As a result, reservoir-induced riverine blooms have become an emerging global environmental issue (Giri, 2021; Reid et al., 2019; Zhi et al., 2024b).

Traditional algal bloom monitoring depends on in situ sampling of chlorophyll-a (Chl-a) and algal cell density at fixed sites. Although

* Corresponding author.

E-mail address: pengqd@iwhr.com (Q. Peng).

<https://doi.org/10.1016/j.srs.2026.100369>

Received 31 October 2025; Received in revised form 15 December 2025; Accepted 6 January 2026

Available online 7 January 2026

2666-0172/© 2026 The Authors. Published by Elsevier B.V. This is an open access article under the CC BY license (<http://creativecommons.org/licenses/by/4.0/>).

accurate, this approach is time-consuming, labor-intensive, and constrained by low sampling frequency, often missing rapidly developing bloom events. Thus, technologies capable of capturing bloom dynamics at high spatiotemporal resolution are urgently needed for effective water management. With advances in remote sensing now provide powerful alternatives, enabling large-scale, continuous observations of bloom evolution. By integrating long-term, high-resolution imagery with field measurements, researchers can monitor algal bloom dynamics more efficiently and comprehensively (Kwon et al., 2020; Shi et al., 2017; Wang and Yang, 2019). High spatial resolution data often have low temporal resolution, limiting their use for long-term monitoring, while high temporal resolution datasets lack fine spatial detail (Yang et al., 2022b). Sentinel-2 provides a 10 m spatial resolution that effectively balances spatial detail and temporal coverage, making it well suited for detecting algal bloom dynamics, even in small tributaries.

High-resolution remote sensing imagery, such as Sentinel-2 scenes (about 1 GB each), requires substantial storage and specialized software for processing. Long-term monitoring of algal bloom dynamics often involves large-scale temporal datasets, presenting challenges in storage, management, and processing. Google Earth Engine (GEE), an open-access cloud computing platform, addresses these challenges by integrating extensive remote sensing data archives and providing efficient, user-friendly programming interfaces (Gorelick et al., 2017). It supports cloud-based data storage, processing, and sharing, enabling large-scale remote sensing analysis and advancing environmental monitoring and scientific research (Yang et al., 2017a).

Machine learning (ML) techniques have been increasingly applied to the screening of bloom-related spectral indices and the retrieval of Chl-a from remote sensing data (Beal et al., 2024; Wang et al., 2025). ML models and their ensemble learning frameworks have demonstrated promising performance across diverse inland waters systems using multisource observations (Li et al., 2021b; Zhang et al., 2024). However, most ML-based approaches require large training datasets that adequately represent diverse optical conditions, which limits their applicability in data-scarce regions (Wang et al., 2025). Random forest (RF) models offer a robust alternative, as they can maintain high predictive accuracy and generalization capability under small-sample conditions. In addition, their intrinsic feature importance evaluation enables efficient screening of informative spectral indices, making RF particularly suitable for algal bloom monitoring in inland waters where in situ observations are limited (Alález et al., 2021; Salman et al., 2024; Wang et al., 2022).

The Three Gorges Reservoir (TGR), the world's largest hydropower project, has significantly altered the hydrological and hydrodynamic conditions of its tributary bays. Backwater effects from the mainstem have caused dramatic reductions in flow velocity. For example, in the Xiangxi Bay (XXB), the average flow velocity during the dry season decreased from 0.73 m/s to 0.01 m/s, and in the Daning River, from 1 to 3 m/s to 0.01–0.08 m/s. These changes, along with increased hydraulic residence time, have worsened water quality issues like eutrophication and algal blooms (Jiang et al., 2025; Li et al., 2022; Zhi et al., 2024a; Zhou et al., 2022). Understanding the spatiotemporal distribution of blooms and their driving factors is essential for managing water quality and ecological security in TGR tributaries.

XXB, the tributary closest to the TGR dam, has experienced frequent algal blooms in recent years (Chuo et al., 2019; Ma et al., 2015; Peng et al., 2023). Although an ecological monitoring station in XXB collects routine water samples, in situ monitoring alone cannot capture bloom dynamics. To address this, this study used Sentinel-2 imagery, hydro-meteorological data, and in situ Chl-a data from 2019 to 2023 to identify bloom distribution and driving mechanisms in XXB. The objectives of this study were to: (1) identify spectral indices for Chl-a and develop accurate retrieval model; (2) analyze monthly Chl-a dynamics, generate high-resolution bloom frequency distributions, and assess bloom intensity; (3) elucidate the hydrometeorological drivers of algal bloom formation in the XXB and propose control strategies.

2. Materials and methods

2.1. Study area

The TGR on the Yangtze River is the largest hydropower project in the world. It is situated between Hubei Province and Chongqing, with an area of approximately 1084 km² at its normal water level of 175 m. The reservoir plays a critical role in flood control, hydropower generation, and water resource management, among other vital functions (Ashour et al., 2021). TGR has been in full operation, and its water level has exhibited regular annual fluctuations between 145 m and 175 m. These fluctuations have caused substantial alterations in the hydrological and aquatic environmental conditions of the reservoir's tributary backwater areas. Hydrodynamic conditions in these regions have significantly weakened, and under the combined influence of increased nutrient concentrations, reduced flow velocity, and so on, major tributaries have experienced algal blooms annually (Zhi et al., 2024b; Zhou et al., 2022). According to monitoring data, the XXB has recorded the highest frequency of algal bloom occurrences among all major tributaries and near-dam areas over the past 13 years (Fig. 1a) (Institute of Hydrobiology Chinese Academy of Sciences, 2024).

The XXB, located in Hubei Province, is the largest tributary closest to the TGR dam site, situated approximately 32 km upstream (Gao et al., 2018). Following the impoundment of the TGR, the XXB has evolved into a transitional waterbody between a river and a lake. Due to reservoir regulation, a backwater zone approximately 25–40 km in length has formed, commonly referred to as XXB (Fig. 1b). Since the impoundment of the TGR, algal blooms of varying intensities have occurred in XXB almost annually. Since 2010, more than nine major bloom events have been recorded, each lasting over seven days and affecting river sections longer than 2 km (Yao et al., 2022). These blooms have had notable impacts on ecosystem health and water quality safety within the TGR. Therefore, the XXB was selected as a representative case study.

2.2. Data collection and processing

Chlorophyll-a (Chl-a) is the primary photosynthetic pigment in phytoplankton, and its concentration reflects the biomass and primary productivity of photosynthetic autotrophs in aquatic ecosystems. It serves as a key indicator for assessing eutrophication and is widely used to characterize the presence of algal blooms. Accordingly, this study employs in situ surface Chl-a as ground-truth data to support the spatially distributed remote sensing retrieval of Chl-a concentrations. Monthly in situ Chl-a data collected from the XXB between 2019 and 2023 were used in this study. The monitoring sites were evenly distributed within XXB, consisting of six sampling points X1 through X6 (Fig. 1b). Specifically, sampling was conducted once per month from 2019 to 2023, with the sampling dates varying each month of each year. The position of each site was recorded using a handheld GPS. Water samples were collected at a depth of 0.5 m below the water surface and in the central area of the bay. Sample collection and analysis followed the national environmental protection standard of the *Water Quality—Determination of Chlorophyll a—Spectrophotometric Method (HJ 897–2017)* (Ministry of Ecology and Environment of the People's Republic of China, 2017). The measurement error of the concentrations met the required precision and accuracy specified in the standard. In this study, the relative standard deviation of the measured Chl-a concentrations ranged from 5.6 % to 11.0 %.

Sentinel-2 imagery from 2019 to 2023, obtained from the Sentinel-2 Surface Reflectance Harmonized (S2_SR_HARMONIZED) dataset on the GEE platform, was used in this study. This dataset is derived from the Copernicus Sentinel-2 Multispectral Instrument (MSI) Level-2A product and provides surface reflectance data. The Sentinel-2 Level-2A surface reflectance products are produced by applying radiometric calibration and atmospheric correction to Level-1C top-of-atmosphere reflectance using the ESA Sen2Cor processor. These harmonized data are widely

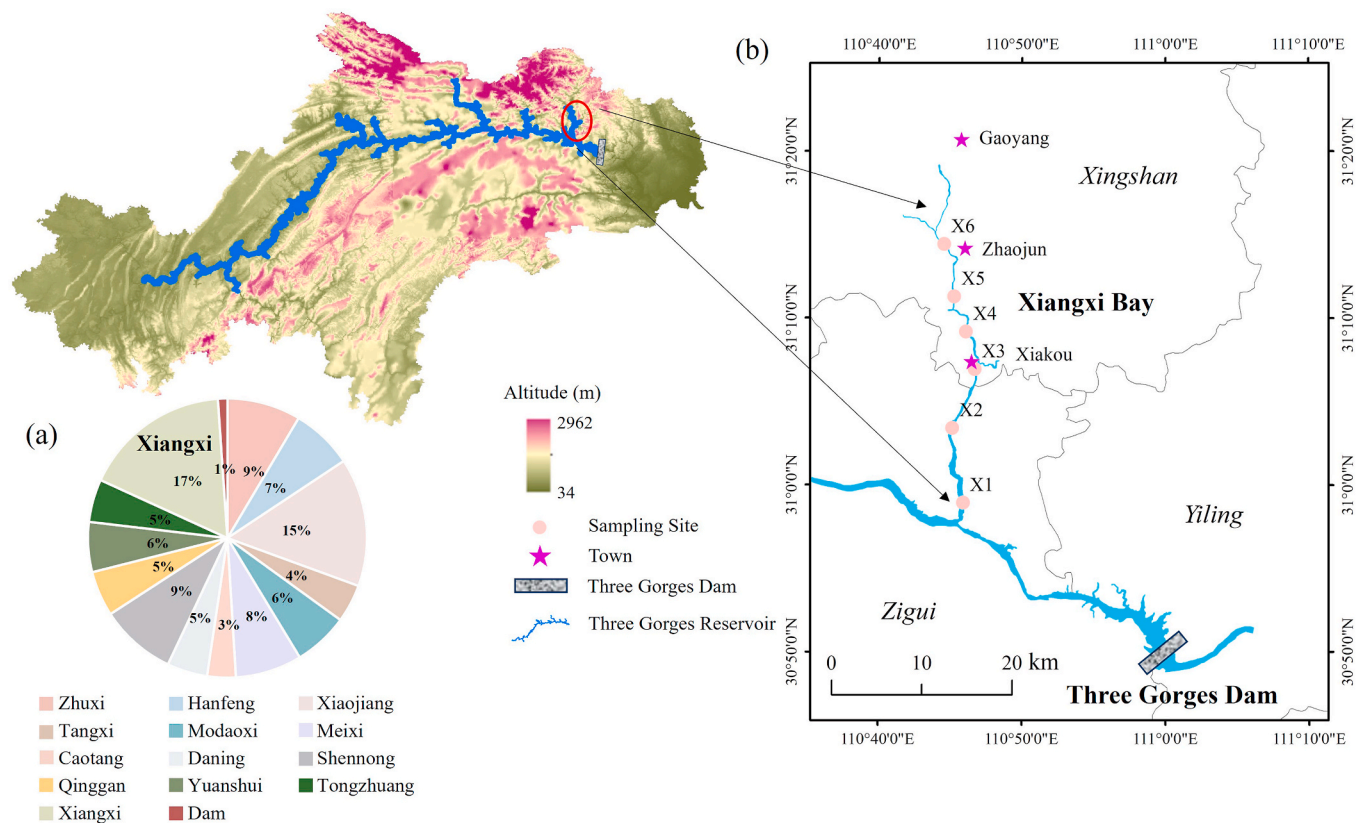


Fig. 1. Overview of the study area. (a) Frequency of algal blooms in major tributaries and dam areas of the TGR during 2010–2023 (Institute of Hydrobiology, Chinese Academy of Sciences, 2024); (b) Study area and sampling points.

applicable to environmental monitoring, including land cover, vegetation, and water quality.

A total of 45 cloud-free Sentinel-2 images covering the study area were acquired for this study. Among them, 12 images correspond to in situ sampling periods, with a maximum temporal difference of three days between satellite acquisition and field measurements. The remaining 35 images capture the algal bloom season, spanning from March to September. The acquisition dates are listed in Table 1.

To enhance image quality and minimize cloud contamination in remote sensing analysis, a cloud masking procedure was applied during the preprocessing stage using Sentinel-2 Level-2A products. Specifically, the S2_SR_HARMONIZED image collection was selected, and its quality assurance band (QA60) was used to identify cloud and cirrus contamination. QA60 is a 16-bit mask band in which the 10th and 11th bits indicate the presence of clouds and cirrus clouds, respectively. Bitwise operations were performed on this band to extract the corresponding binary flags. A pixel-level mask was then constructed, and only pixels with both bits equal to zero were considered cloud-free and retained for further analysis.

Furthermore, to standardize the data scale, the original surface reflectance values (ranging from 0 to 10,000) were normalized to a 0–1 range. The cloud masking function was applied across the entire image collection to systematically eliminate cloud-contaminated areas, thereby ensuring the accuracy of subsequent water body monitoring and remote sensing index calculations. Given that Sentinel-2 bands have spatial resolutions ranging from 10 to 60 m, all selected images were resampled to a uniform spatial resolution of 10 m prior to remote sensing index calculation.

Based on a comprehensive review of the literature, thirteen spectral indices associated with Chl-a were selected for further analysis. These include the Normalized Difference Chlorophyll Index (NDCI), Maximum

Table 1
Summary of Sentinel-2 images collected for the study area.

Year	2019	2020	2021	2022	2023
Sentinel-2 remote sensing images date	2019-4-1 2019-4-6★ 2019-8-14 2019-8-19 2019-8-29★ 2019-9-23 2019-9-28★ 2019-10-18	2020-3-1 2020-4-15 2020-5-15 2020-6-4 2020-6-24 2020-7-9 2020-8-3 2020-8-28	2021-4-30 2021-6-24 2021-7-14 2021-7-29 2021-9-12 2021-9-22 2021-10-2 2021-10-22	2022-2-4★ 2022-2-24★ 2022-3-11★ 2022-4-10★ 2022-5-10★ 2022-7-24 2022-8-3 2022-8-13 2022-9-12★ 2022-10-12★ 2022-10-22 2022-11-1★	2023-4-15 2023-4-20 2023-4-25 2023-4-30 2023-8-23 2023-10-22★ 2023-12-6★ 2023-12-6★
Total	8	11	7	12	7
Sampling period (up to 3 days off)	3	0	0	7	2
Algal bloom period (March–September)	7	10	6	7	5

Note: ★ Remote sensing images acquired during the in situ sampling period or within a maximum of three days of the sampling date.

Chlorophyll Index (MCI), Algal Bloom Detection Index (ABDI), Normalized Difference Turbidity Index (NDTI), Floating Algae Index (FAI), Normalized Difference Vegetation Index (NDVI), as well as several band ratio indices: B8AB4, B3B2, B8B3, B8B4, B6B5, B3B4, and B5B4. The formulas for calculating these indices are listed in Table 2, and the corresponding Sentinel-2 spectral bands are detailed in Table 3.

2.3. Analytical method

All Sentinel-2 image selection, cloud masking, normalization, resampling, and spectral index calculations were performed on the GEE platform (Fig. 2A). The extracted feature index values at the in situ sampling locations were subsequently exported to support the development of Chl-a retrieval models. The overall technical workflow is presented in Fig. 2.

Details of the remote sensing data processing are presented in Section 2.2. After preprocessing on the GEE platform, values of spectral indices at the sampling points were calculated and exported. Chl-a concentrations measured in situ during the sampling periods were used as the dependent variable, while the corresponding thirteen spectral indices (acquired within ± 3 days of the sampling dates) served as independent variables. The RF model was employed for feature selection based on permutation importance (Fig. 2B). The importance of variables was ranked using the “RandomForest” package in R (Liaw and Wiener, 2007). The model was trained on the first 80 % of the temporal dataset, and the significance of variable importance was assessed through 1000 permutations using the “rfPermute” package. To enhance the stability of the variable importance estimates under the constraint of a limited sample size, the model was evaluated using 50 repetitions of 5-fold cross-validation. Variable importance scores were obtained for each run, and the final importance values were calculated as the mean across all repeated folds. In RF variable importance is assessed using two indices: the percentage increase in mean squared error (%IncMSE) and the increase in node purity (IncNodePurity). %IncMSE reflects the effect on predictive accuracy, while IncNodePurity indicates contributions to tree splits. Variables showing high values in both indices were identified as key drivers of the model's predictions (Sang et al., 2024; Zheng et al., 2020).

Table 2
Sentinel-2 spectral indices used for Chl-a feature index.

Index	Formula	References
NDCI	$(B5 - B4)/(B5 + B4)$	(Kislik et al., 2022; Mishra and Mishra, 2012; Zhi et al., 2024a)
MCI	$(B8A - B4) - 0.533 \times (B6 - B4)$	(Gower et al., 2008; Zhang et al., 2024)
ABDI	$[B6 - B4 - (B8A - B4) \times (\lambda_{RE2} - \lambda_{Red}) / (\lambda_{NIRn} - \lambda_{Red})] - (B4 - 0.5 \times R_{Green})$	Qing et al. (2021)
NDTI	$(B4 - B3)/(B4 + B3)$	(Elhag et al., 2019; Zhang et al., 2024)
FAI	$B8 - [B4 + (B11 - B4) \times (\lambda_{NIR} - \lambda_{Red}) / (\lambda_{SWIR} - \lambda_{Red})]$	(Liu et al., 2021; Qing et al., 2021)
NDVI	$(B8 - B4)/(B8 + B4)$	Ma et al. (2021)
B8AB4	$(B8A - B4)/(B8A + B4)$	(Khalili and Hasanlou, 2019; Kislik et al., 2022)
B3B2	$(B3 - B2)/(B3 + B2)$	Kislik et al. (2022)
B8B3	$B8/B3$	(Hossen et al., 2022; Zhang et al., 2024)
B8B4	$B8/B4$	(Yang et al., 2017; Zhang et al., 2024)
B6B5	$B6/B5$	(Wang et al., 2021; Zhang et al., 2024)
B3B4	$B3/B4$	(Kim et al., 2022; Zhang et al., 2024)
B5B4	$B5/B4$	(Shi et al., 2022; Zhang et al., 2024)

Table 3

Description of Sentinel-2 spectral bands.

Band	Name	Central wavelength (nm)	Spatial resolution (m)
B1	Coastal aerosol	443	60
B2	Blue	490	10
B3	Green	560	10
B4	Red	665	10
B5	Vegetation red edge 1	705	20
B6	Vegetation red edge 2	740	20
B7	Vegetation red edge 3	783	20
B8	Near Infrared	842	10
B8A	Narrow NIR (Red Edge 4)	865	20
B9	Water vapor	945	60
B10	SWIR – Cirrus	1375	60
B11	SWIR 1	1610	20
B12	SWIR 2	2190	20

Subsequently, polynomial regression models were constructed using the selected Chl-a spectral indices and corresponding in situ measurements (Fig. 2B), with remote sensing image acquisition dates restricted to within ± 3 days of the sampling dates. The model exhibiting the highest goodness-of-fit was then employed to retrieve Chl-a concentrations during the algal bloom seasons (March to September) from 2019 to 2023. Standardized anomalies were further calculated to assess potential algal bloom occurrences across different months (Kislik et al., 2022) (Fig. 2D). The standardized anomaly was computed as follows:

$$Z_{M-abnormal} = \frac{(Chla_{Month} - Chla_{Mean})}{Chla_{STD}}$$

In the formula, $Z_{M-abnormal}$ denotes the anomaly of Chl-a during the algal bloom season. $Chla_{Month}$ represents the Chl-a concentration for each month, while $Chla_{Mean}$ is the mean Chl-a concentration averaged over multiple months (March to September). $Chla_{STD}$ refers to the standard deviation of Chl-a concentrations across these months. A value of $Z_{M-abnormal} > 0$ indicates a positive anomaly, meaning Chl-a concentrations exceed the long-term average, with larger values reflecting stronger deviations. Conversely, $Z_{M-abnormal} < 0$ signifies a negative anomaly, where concentrations are below the long-term average, and more negative values correspond to more pronounced anomalies. $Z_{M-abnormal} \approx 0$ indicate conditions near the long-term average.

In this study, the Chl-a threshold method was employed to identify algal bloom. Based on previous studies (Li et al., 2021a; Zheng et al., 2006) and recent specialized monitoring of algal blooms in the XXB, a critical Chl-a concentration threshold of 47.7 $\mu\text{g/L}$ was used to determine bloom occurrence (Fig. 2C). The frequency of algal bloom was calculated using the following formula:

$$F = \frac{N_{bloom}}{M} \times 100\%$$

In the formula, F denotes the frequency of algal bloom occurrence (%), N_{bloom} represents the number of bloom events, and M represents the total number of observations considered, which in this study corresponds to the number of analyzed images.

Algal bloom intensity reflects the overall severity of blooms within the study area. This study follows the technical specification for algal bloom evaluation through the standard of *Technical Specification for Algal Bloom Remote Sensing and Ground Monitoring Evaluation (Trial)* (HJ 1098–2020) (Ministry of Ecology and Environment of the People's Republic of China, 2020) (Fig. 2D). The classification of algal bloom intensity is based on the proportion of the area affected by blooms, and the grading criteria are summarized in Table 4.

The driving factors of algal blooms in XXB were investigated from two perspectives. First, external influences were assessed using box plots and the Mann–Whitney U test to compare bloom and non-bloom conditions across nine hydrometeorological variables include TGR inflow

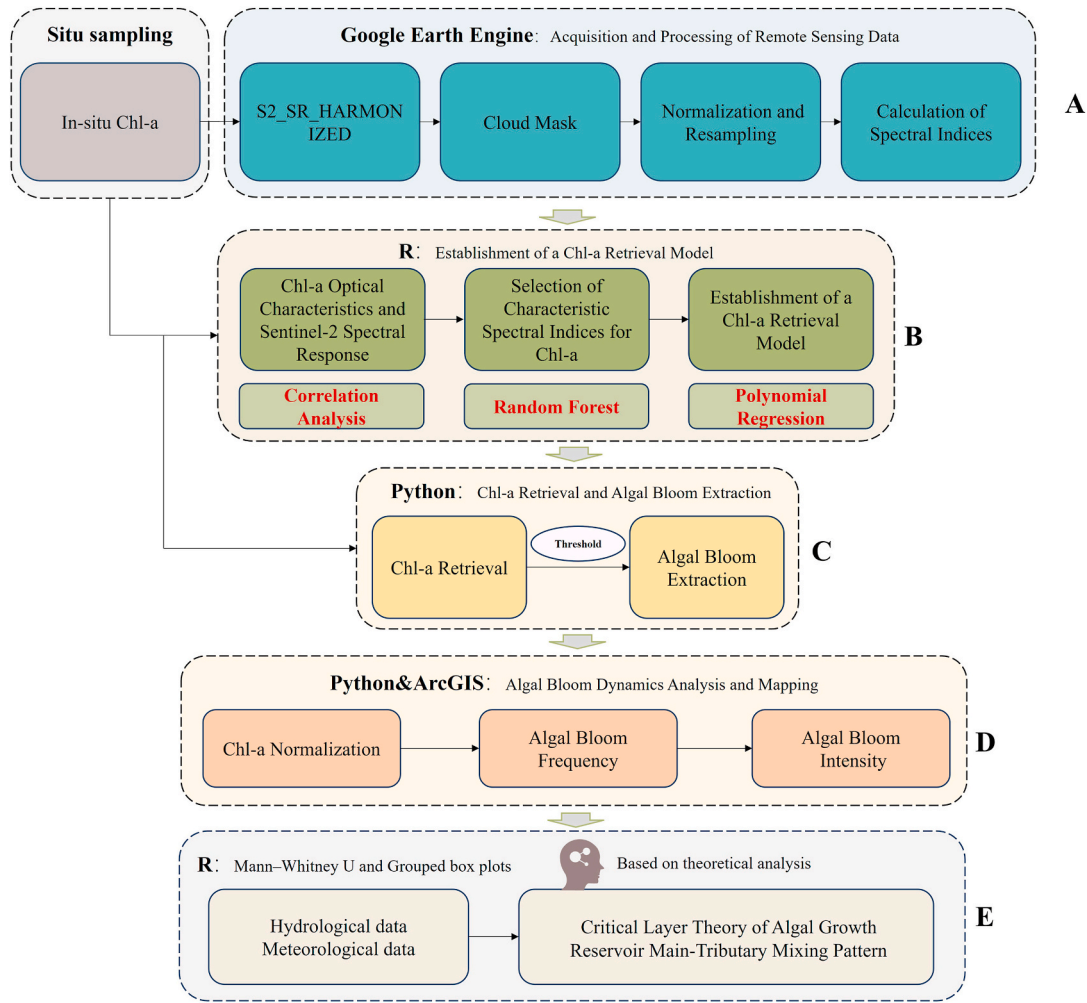


Fig. 2. Schematic diagram of the technical workflow.

Table 4
Grading criteria for algal bloom intensity.

Algal bloom intensity levels	Percentage of algal bloom coverage P(%)	Algal bloom characteristics	Characteristic phenomenon
I	0	No bloom	No obvious algal bloom observed on the water surface
II	$0 < P < 10$	No obvious bloom	Scattered algal blooms observed on the water surface
III	$10 \leq P < 30$	Mild bloom	Localized algal blooms observed on the water surface
IV	$30 \leq P < 60$	Moderate bloom	Regional algal blooms observed on the water surface
V	$60 \leq P \leq 100$	Severe bloom	Widespread algal blooms observed on the water surface

(TGRQ), Xiangshan Station inflow (XSQ), TGR water level (TGRWL), Xiangshan Station water level (XSWL), main-tributary water level difference (WLD), the range of water level variation in the TGR (TGRWLTV) and the range of water level variation in the Xingshan (XSWLTV), 10-day cumulative rainfall (RAIN10D), and wind speed (WIND). Second, internal mechanisms were examined by integrating the critical depth theory of algal growth with the hydrodynamic mixing regime of XXB

(Fig. 2E).

The selection and preprocessing of remote sensing imagery—including large-scale scene filtering and the calculation of spectral indices—were conducted on the GEE platform, which enabled efficient handling of massive datasets and minimized the computational burden on local environments. Subsequent analytical tasks were implemented using a modular workflow in which each software environment was assigned to the operations it performs best. Feature selection for Chl-a retrieval and the development of Chl-a prediction models were carried out in R 4.3.3, which offers rich statistical packages and flexible visualization capabilities. Chl-a inversion and algal bloom extraction were implemented in Python 3.12.9, while analyses of bloom dynamics and map production were completed using Python 3.12.9 together with ArcGIS 10.6, which enables pixel-level inspection and high-quality cartographic outputs. Boxplot generation and the nonparametric Mann-Whitney U tests were also performed in R 4.3.3. This modular design ensures that each processing step leverages the strengths of the respective tools, thereby improving both computational efficiency and analytical reliability.

3. Results

3.1. Chl-a retrieval in the XXB using remote sensing

3.1.1. Chl-a optical characteristics and spectral response

Phytoplankton contain a variety of photosynthetic pigments, with Chl-a serving as the primary pigment driving photosynthesis. The

concentration of Chl-a in water is a key indicator of nutrient status and an essential parameter for characterizing algal blooms. The molecular structure of Chl-a determines its unique optical properties, providing the physical basis for its retrieval via remote sensing. Chl-a strongly absorbs light at specific wavelengths and emits fluorescence during photosynthesis, producing distinctive spectral features in water reflectance that can be detected.

Chl-a absorbs most of its energy in the blue and red-orange regions of the spectrum, converting light energy into chemical energy, while primarily reflecting green light. Fig. 3a shows the absorption spectra of Chl-a and Chl-b in typical water bodies. Numerous studies have demonstrated that an increase in Chl-a concentration leads to a reduction in spectral response at shorter wavelengths, particularly in the blue region, while strong absorption occurs in the red region, producing characteristic peaks and troughs associated with algal presence (Gholizadeh et al., 2016; Lichtenthaler, 1987).

Chl-a, together with suspended particulate matter and colored dissolved organic matter, constitutes the primary optically active substances in aquatic systems. Owing to differences in sources, composition, and terrestrial inputs, inland waters display substantial regional variability in their inherent optical properties. Fig. 3b shows the spectral reflectance curves of the main Sentinel-2 bands for the XXB. The reflectance displays pronounced peaks in the Blue (B2) and Green (B3) bands, while in the red-edge bands B5 (around 705 nm) and B7 (around 783 nm), the reflectance shows a sudden increase with relatively weak peaks and troughs. Further analysis of the relationship between Chl-a concentration and the main Sentinel-2 bands (B2–B8A) is presented in Fig. 3c and d. To improve the consistency and correlation between remote sensing data and in situ measurements, satellite images were restricted to those acquired within three days of field sampling. Images exhibiting pronounced turbidity due to heavy rainfall or those contaminated by cloud cover were excluded. Ultimately, 42 matched

sample pairs were obtained for correlation analysis. Chl-a exhibits positive linear correlations with B5 ($r = 0.42$), B6 ($r = 0.35$), B7 ($r = 0.33$), B8A ($r = 0.32$), and B8 ($r = 0.24$), and a strong non-linear correlation with B2 (MIC = 0.99). High inter-band correlations, particularly linear correlations, were also observed (Fig. 3c). Collectively, these results indicate that Chl-a in the XXB can be effectively retrieved using Sentinel-2 imagery, facilitating the monitoring of algal bloom dynamics.

3.1.2. Selection of spectral indices

Based on the analysis in Section 3.1.1, Chl-a exhibits both linear and non-linear correlations with individual Sentinel-2 bands. Many researchers have proposed a variety of Chl-a retrieval indices, often derived from individual bands or constructed through different band combinations. In this study, measured Chl-a concentrations at sampling sites were used as the dependent variable, while thirteen spectral indices (NDCI, MCI, ABDI, NDTI, FAI, NDVI, B8AB4, B3B2, B8B3, B8B4, B6B5, B3B4, and B5B4) calculated for each site with a maximum temporal offset of three days, were selected as independent variables. The RF model was then applied to establish the Chl-a prediction model. This approach aimed to identify spectral indices that significantly influence Chl-a concentrations, thereby providing robust characteristic indices for model construction and enhancing the accuracy and efficiency of Chl-a retrieval.

In this study, the RF model was applied to identify the most important spectral indices associated with Chl-a. The model results indicated that, compared to other indices, MCI, ABDI, NDCI, and B5B4 exhibited relatively high values in both %IncMSE and IncNodePurity. Specifically, regarding %IncMSE, the indices ranked as follows: B5B4 > NDCI > ABDI > MCI; whereas for IncNodePurity, the ranking was: ABDI > B5B4 > NDCI > MCI (Fig. 4). These results suggest that these four spectral indices play a critical role in Chl-a prediction and can be considered key characteristic indices for Chl-a retrieval.

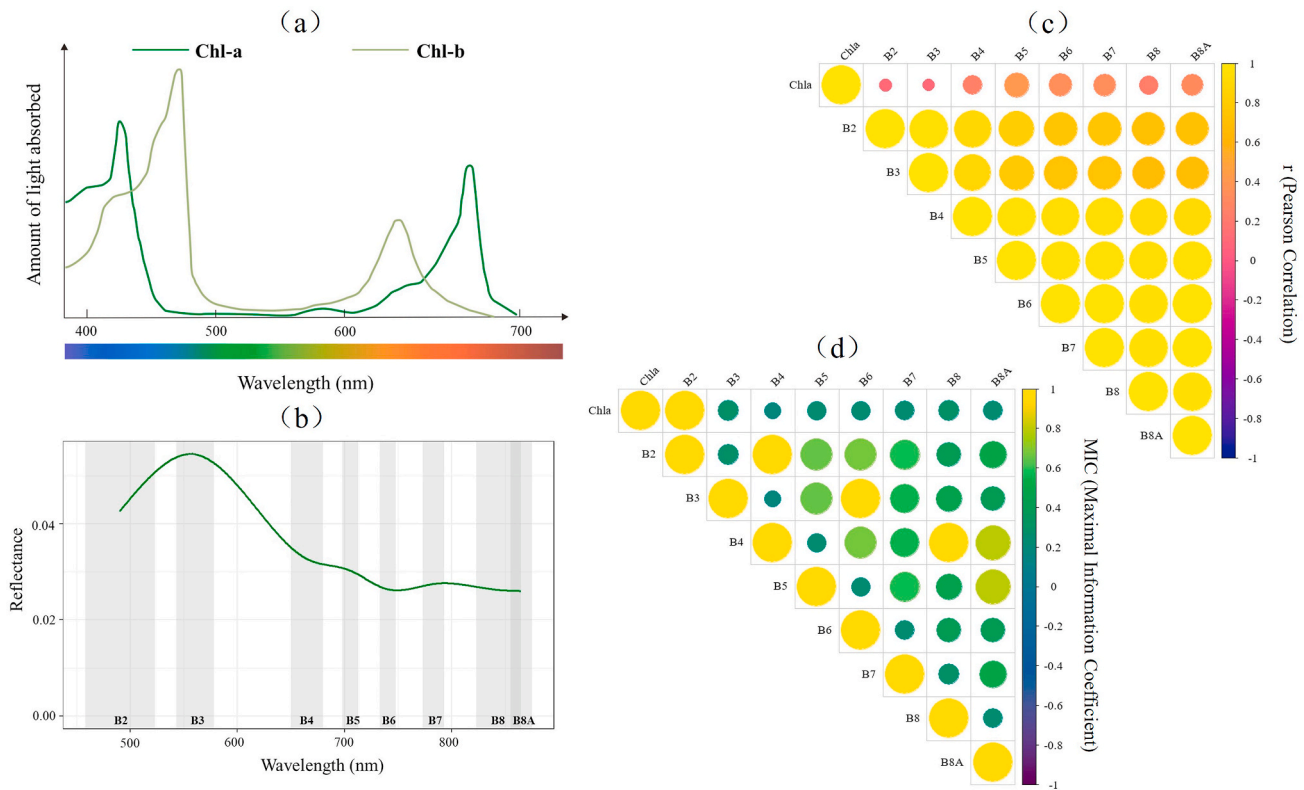


Fig. 3. Spectral characteristics of Chl-a and its correlation with Sentinel-2 bands

(a) The absorption spectra of Chl-a and Chl-b in typical water bodies (Gholizadeh et al., 2016; Lichtenthaler, 1987); (b) The spectral reflectance curves of the main Sentinel-2 bands for the XXB; (c) and (d) Heatmap of correlations between Chl-a and the main Sentinel-2 bands.

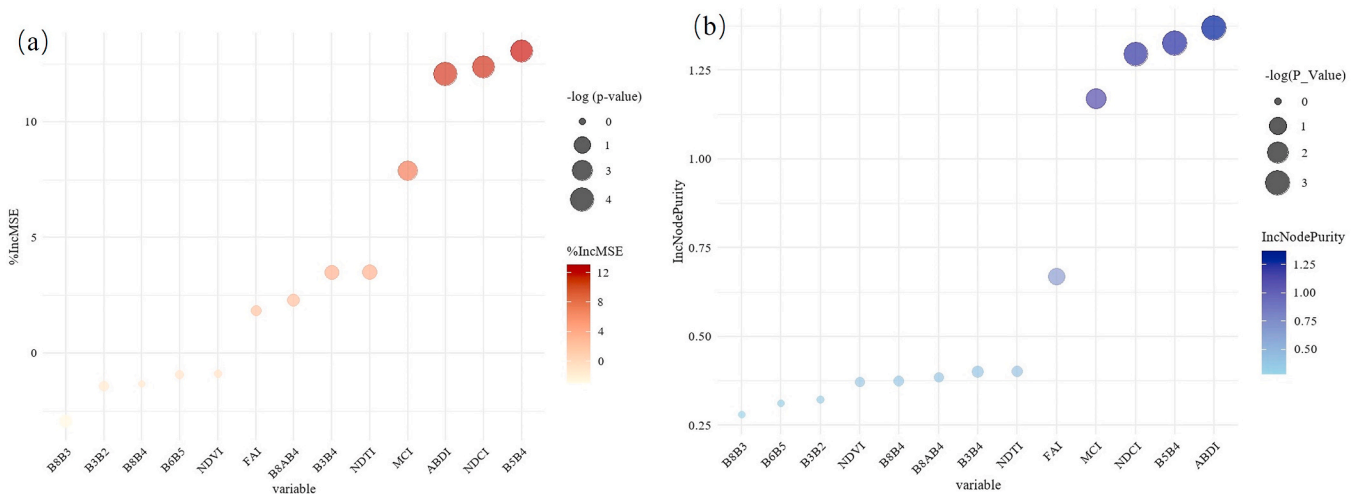


Fig. 4. Importance of spectral indices in Chl-a prediction using RF model.

3.1.3. Chl-a retrieval models

Based on the analysis in Section 3.1.1, Chl-a exhibited both linear and nonlinear correlations with individual Sentinel-2 bands, while the bands also showed strong inter-correlations (Fig. 3c and d). Numerous studies have utilized single bands as the basis for constructing various Chl-a retrieval indices through different band combinations. In this study, we constructed polynomial regression models for Chl-a retrieval (Fig. 5), using measured Chl-a concentrations at the sampling sites as the dependent variable and four bloom-related spectral indices (MCI, ABDI, B5B4, and NDCI)—selected through the RF model—as independent variables. To improve the predictive performance of the regression

models, spectral indices were derived from remote sensing images acquired within three days of field sampling. Images exhibiting pronounced turbidity due to heavy rainfall or contaminated by cloud cover were excluded. Ultimately, 42 matched sample pairs (in situ Chl-a and remote sensing indices) were used to develop the regression models. Among these indices, the NDCI-based model achieved the highest fitting performance ($R^2 = 0.76$), followed closely by the B5B4-based model ($R^2 = 0.75$).

The superior performance of NDCI can be attributed to its normalization mechanism, defined as $(B5 - B4)/(B5 + B4)$, which effectively mitigates interference caused by variations in solar zenith angle, aerosol

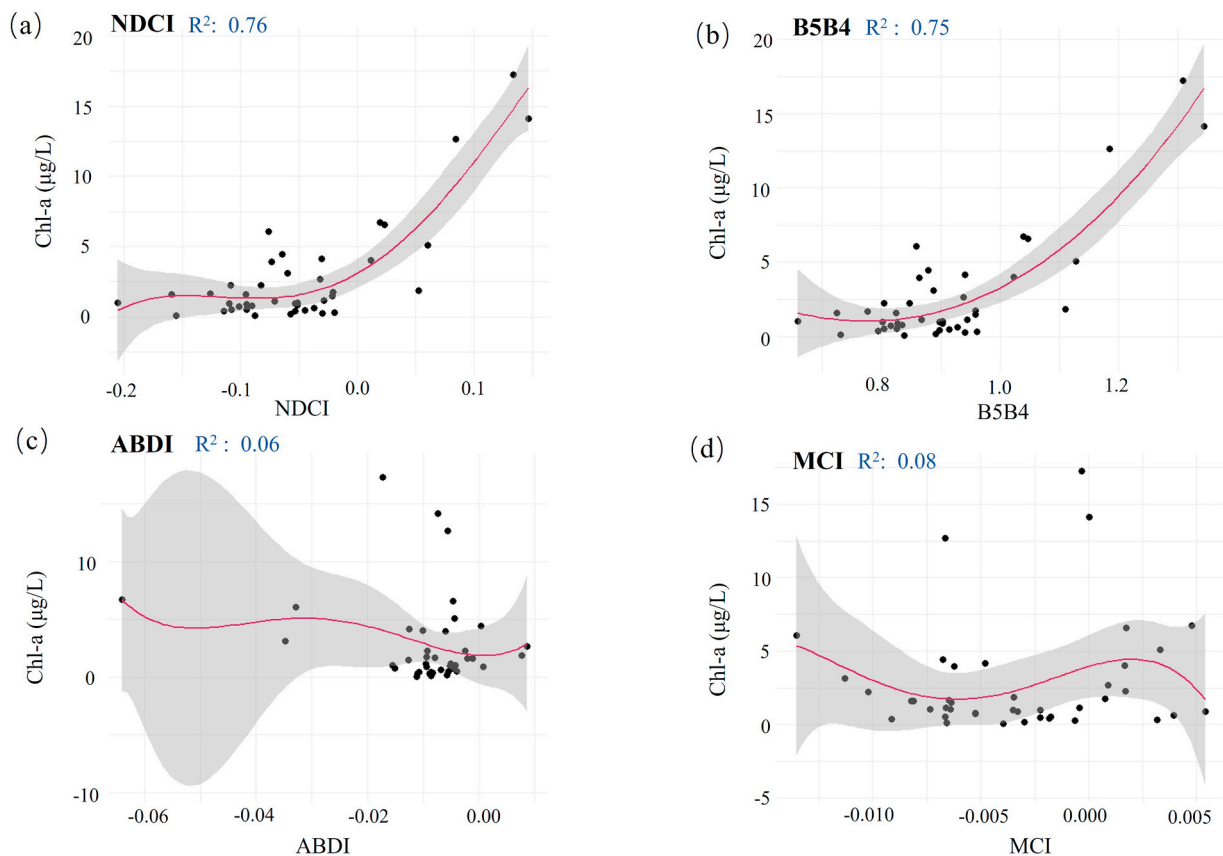


Fig. 5. Polynomial regression relationships between four spectral indices (NDCI, B5B4, ABDI, and MCI) and in situ Chl-a.

content, and atmospheric scattering. In contrast, the simple band ratio B5/B4 is more susceptible to instability, especially when the denominator (B4) is very small or anomalous, potentially leading to exaggerated ratio values and misinterpretation. Furthermore, the NDCI index was originally proposed for estimation of Chl-a concentrations in inland freshwater lakes and eutrophic coastal waters, providing an effective tool for monitoring eutrophication and algal bloom dynamics (Mishra and Mishra, 2012). Since then, NDCI has been widely applied to inland freshwater lakes and large rivers for cyanobacterial bloom assessments, consistently demonstrating stable and robust performance in retrieving Chl-a concentrations (Caballero and Gabriel, 2021).

Therefore, in this study, the polynomial regression model based on the best-performing NDCI index was ultimately selected for Chl-a retrieval. The final regression equation is as follows:

$$\text{Chl-a} = 3.1658 + 40.5537 \times \text{NDCI} + 269.7162 \times \text{NDCI}^2 + 592.2277 \times \text{NDCI}^3$$

This model was applied to retrieve Chl-a concentrations from 35 satellite images acquired during bloom periods, enabling further analysis of bloom dynamics.

Based on the optimal polynomial regression model, an automated Chl-a retrieval was developed in Python 3.12.9. Fig. 6 shows selected retrieval results, which depict the spatiotemporal distribution of Chl-a in the XXB on representative dates during bloom-prone periods. As shown, higher Chl-a levels are concentrated in the mid-to upper reaches of XXB, which have historically exhibited frequent algal blooms. This spatial pattern shows good agreement with long-term in situ water quality monitoring data from the XXB (Luo et al., 2022). The combined influence of intensive anthropogenic activities and distinctive hydrodynamic

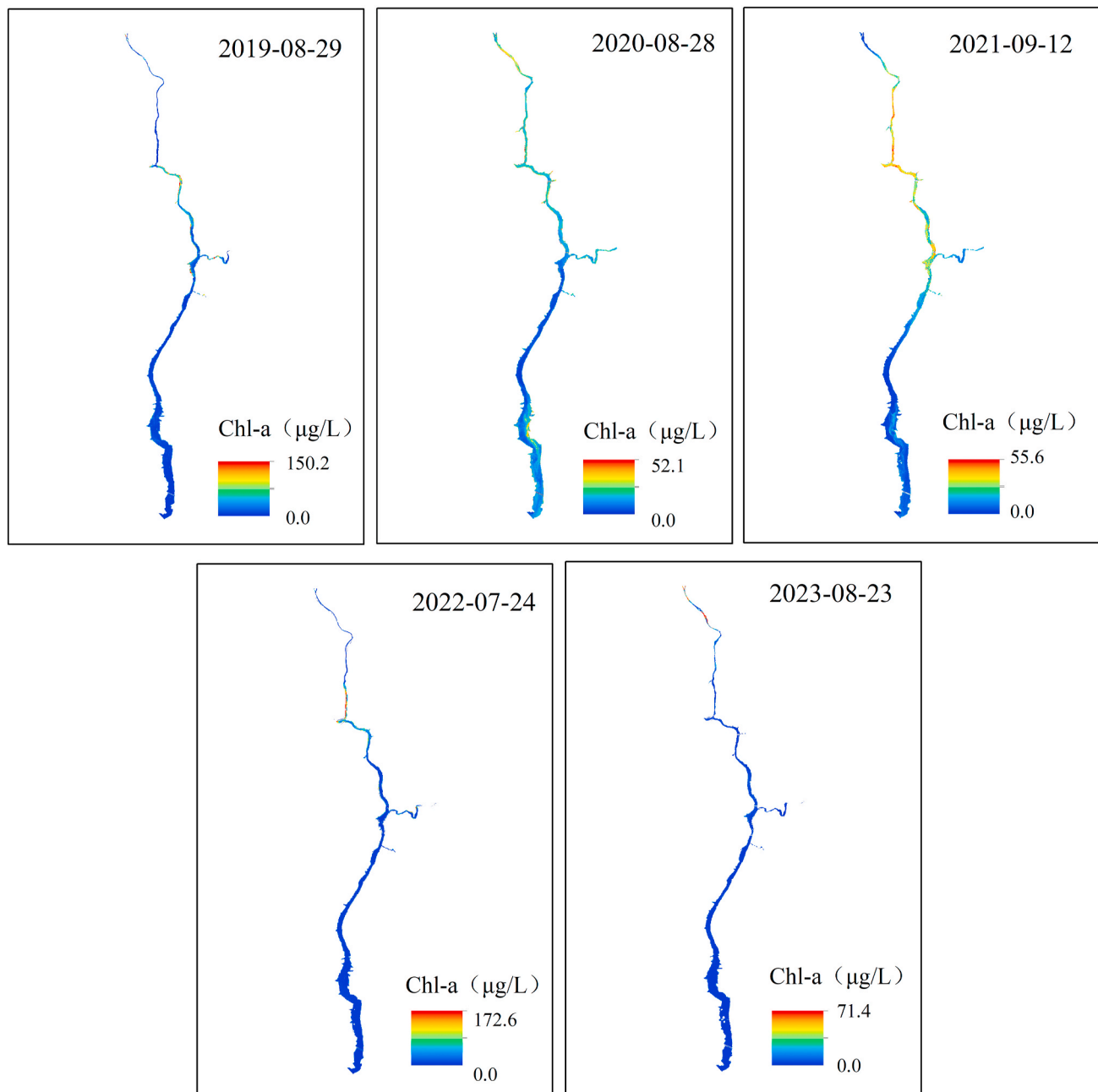


Fig. 6. Chl-a distribution maps retrieved for representative dates during bloom-prone periods in the XXB.

conditions in this region facilitates nutrient accumulation. Furthermore, the reservoir's backwater effect weakens hydrodynamic forces, thereby creating favorable conditions for algal proliferation and making this zone highly susceptible to bloom outbreaks (Jin et al., 2018).

3.2. Spatiotemporal detection of algal blooms

The monthly standardized anomalies of Chl-a reflect the relative spatial and temporal deviations from the long-term average Chl-a concentrations. In this study, these anomalies were calculated for the bloom-prone period (March to September) from 2019 to 2023 (Fig. 7). They represent the difference between the mean Chl-a concentration for

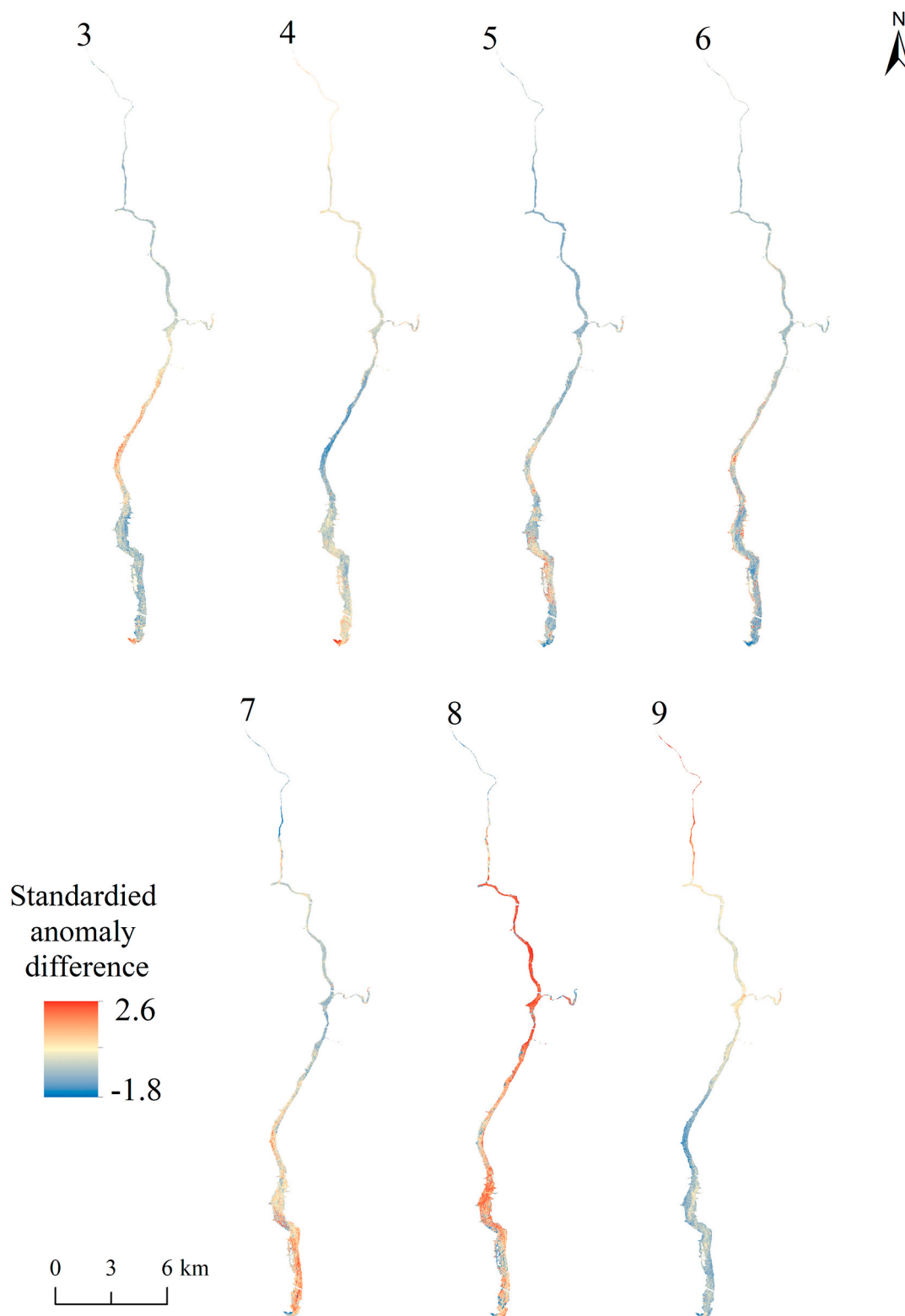


Fig. 7. Spatial distribution of monthly standardized Chl-a concentration anomalies during bloom periods.

each specific month and the long-term average during bloom periods. Positive anomaly values indicate that the monthly average Chl-a concentration exceeds the long-term mean, with larger values corresponding to more significant deviations, which may indicate a potential of algal blooms.

Based on the monthly standardized Chl-a anomaly maps (Fig. 7), it is evident that during the March to June, Chl-a concentrations in the middle and lower region of the XXB were generally higher than those in the upstream region. Specifically, in March, the area extending from Xiakou Town to approximately 7 km downstream exhibited Chl-a levels significantly above the bloom season average. In July, the anomaly zone extended from the bay's estuary to approximately 10 km upstream, exhibiting elevated Chl-a concentrations relative to the mean concentration during the bloom period. In August, high Chl-a anomalies covered the most extensive area across the bay, with nearly 80 % of the surface exhibiting concentrations exceeding the bloom-period average, especially between 5 km downstream and 8 km upstream of Xiakou Town. In September, significant high anomaly zones were primarily distributed from Zhaojun Town to approximately 5 km downstream.

In summary, algal blooms in XXB were most likely to occur in March, July, August, and September in recent years, especially in August.

3.3. Frequency and intensity of algal blooms in the XXB

Algal bloom frequency refers to the proportion of time during the observation period when algal blooms occurred. It is an important indicator of algal bloom dynamics and serves as a key metric for evaluating aquatic ecosystem health and informing water management strategies. In this study, a Chl-a threshold of 47.7 $\mu\text{g/L}$, established from previous monitoring data in the XXB, was employed to identify algal bloom (Li et al., 2021a; Zheng et al., 2006).

Based on 35 remote sensing images acquired during bloom periods (March to September) from 2019 to 2023, the frequency of algal bloom occurrences in the XXB was statistically analyzed. The results indicate that approximately 18 % of the XXB area experienced algal blooms in recent years (Fig. 8a and b). Notably, the upstream region (sites X4–X6) exhibited relatively high bloom frequencies, particularly near Pingyikou (X5), where bloom occurrence exceeded 10 %, reaching a maximum of 23 % (Fig. 8a).

Areas with bloom frequencies between 10 % and 23 % were relatively limited, covering approximately 0.13 km^2 , which accounts for about 2 % of the bloom-affected area (Fig. 8b and c). Regions with bloom frequencies ranging from 0 % to 5 % comprised roughly 9 % of the bloom-affected area, approximately 0.89 km^2 , primarily located between upstream sites X4 and X5 as well as X5 and X6 (Fig. 8).

The intensity of algal blooms in the XXB from 2019 to 2023 was analyzed based on the proportional area affected by blooms. The results indicate that the majority of blooms during this period were Level II, characterized by scattered algal blooms observed on the water surface, accounting for 63 % of the total area (Fig. 9e). Years with relatively higher bloom coverage were primarily concentrated in 2022. Notably, the most intense Level III bloom occurred on August 3, 2022, with bloom coverage reaching 10.62 % (Fig. 9a). This event was predominantly distributed in the upstream region between sites X4 and X5 (Fig. 9d), where the maximum NDCI value reached 0.63 and the retrieved Chl-a concentration peaked at 257.87 $\mu\text{g/L}$ (Fig. 9b and c). Corresponding remote sensing images revealed a bright green coloration in this area (Fig. 9f).

Field measurements on August 9, 2022, confirmed Chl-a concentrations of 238.5 $\mu\text{g/L}$ at site X4 and 29.57 $\mu\text{g/L}$ at site X5. These observations suggest that the bloom dynamics during this period exhibited progression from the upstream to midstream sections.

3.4. Hydrometeorological factors on algal blooms

According to the analyses in Sections 3.2 and 3.3, algal blooms in

recent years have predominantly occurred during the flood season (mid-June to mid-September), mainly distributed in the mid-to upper reaches of XXB. To further clarify the hydrometeorological driving mechanisms underlying algal bloom occurrences, we applied a non-parametric test (Mann–Whitney U) and constructed boxplots of hydrometeorological factors under bloom and non-bloom conditions to explore the potential influence of external hydrometeorological drivers on algal blooms in XXB. The results indicated that both the inflow discharge of the TGR and the upstream discharge of XXB differed significantly between bloom and non-bloom conditions ($p < 0.01$). Blooms occurred mainly under low-flow conditions (Fig. 10a and b). During low-discharge periods, reduced flow velocity and weakened hydrodynamic disturbances created a stable environment favorable for algal bloom development. Regarding water level variation in the TGR and at the Kingshan Station, smaller variations generally provided favorable conditions for algal growth and outbreak, whereas larger variations were unfavorable for bloom formation (Fig. 10f and g). Among the meteorological factors, the 10-day cumulative rainfall also exhibited significant differences between bloom and non-bloom, with blooms tending to occur under relatively low rainfall (Fig. 10h). With the exception of unusual cases, blooms were typically observed when the cumulative rainfall over 10 consecutive days was less than 35 mm. Such limited rainfall supplied less discharge to the river channel, thereby sustaining a relatively stable environment conducive to algal growth.

4. Discussion

4.1. Effectiveness of monitoring algal blooms in narrow tributaries

GEE is a cloud-based remote sensing platform that integrates vast archives of satellite and geospatial data. Freely accessible to users, it enables cloud-based data access, processing, and sharing without the constraints of local memory or computational capacity. Its high-performance computing capabilities and flexible programming interfaces offer effective technical support for dynamic water monitoring, water quality parameter retrieval, and algal bloom detection. In recent years, GEE has shown considerable potential in advancing algal bloom monitoring applications.

Weber et al. (2020) employed the GEE platform to analyze landscape characteristics in the Georgia Piedmont region, investigating the risk of cyanobacterial harmful algal blooms resulting from the interplay between land management practices and climate conditions (Weber et al., 2020). Lucia Lobo et al. (2021) leveraged the GEE platform to develop AlgaeMap, an algal bloom monitoring application based on NDCI and Trophic State Index (TSI). As the first online platform built within GEE, AlgaeMap provides high-spatial-resolution water quality information. These studies highlight the advantages of GEE's long-term data archives, large-scale integration of remote sensing imagery, and efficient capabilities for data processing and sharing (Lobo et al., 2021). By leveraging GEE for data integration and cloud-based processing, this study performed a spatiotemporal analysis of algal blooms in the XXB during 2019–2023, thereby improving data processing efficiency.

Compared with other imagery sources such as Sentinel-1, Landsat, MODIS, Sentinel-2 offers significant advantages in spectral, spatial, and temporal resolution for algal bloom monitoring. In terms of spectral resolution, Sentinel-2 provides comprehensive multispectral coverage, including unique red-edge and shortwave infrared (SWIR) bands. The Multispectral Instrument (MSI) onboard Sentinel-2 features 13 spectral bands spanning the visible, near-infrared, and SWIR regions, notably including four red-edge bands. Cyanobacteria and other bloom-forming organisms exhibit distinctive spectral reflectance characteristics—such as peaks in the red-edge region—which enable more accurate bloom detection through combinations of red-edge and other bands. This approach effectively minimizes interference from suspended solids, sediments, and other optically active water constituents (Toming et al., 2016; Yang et al., 2022a; Zhang et al., 2024). Accordingly, NDCI was

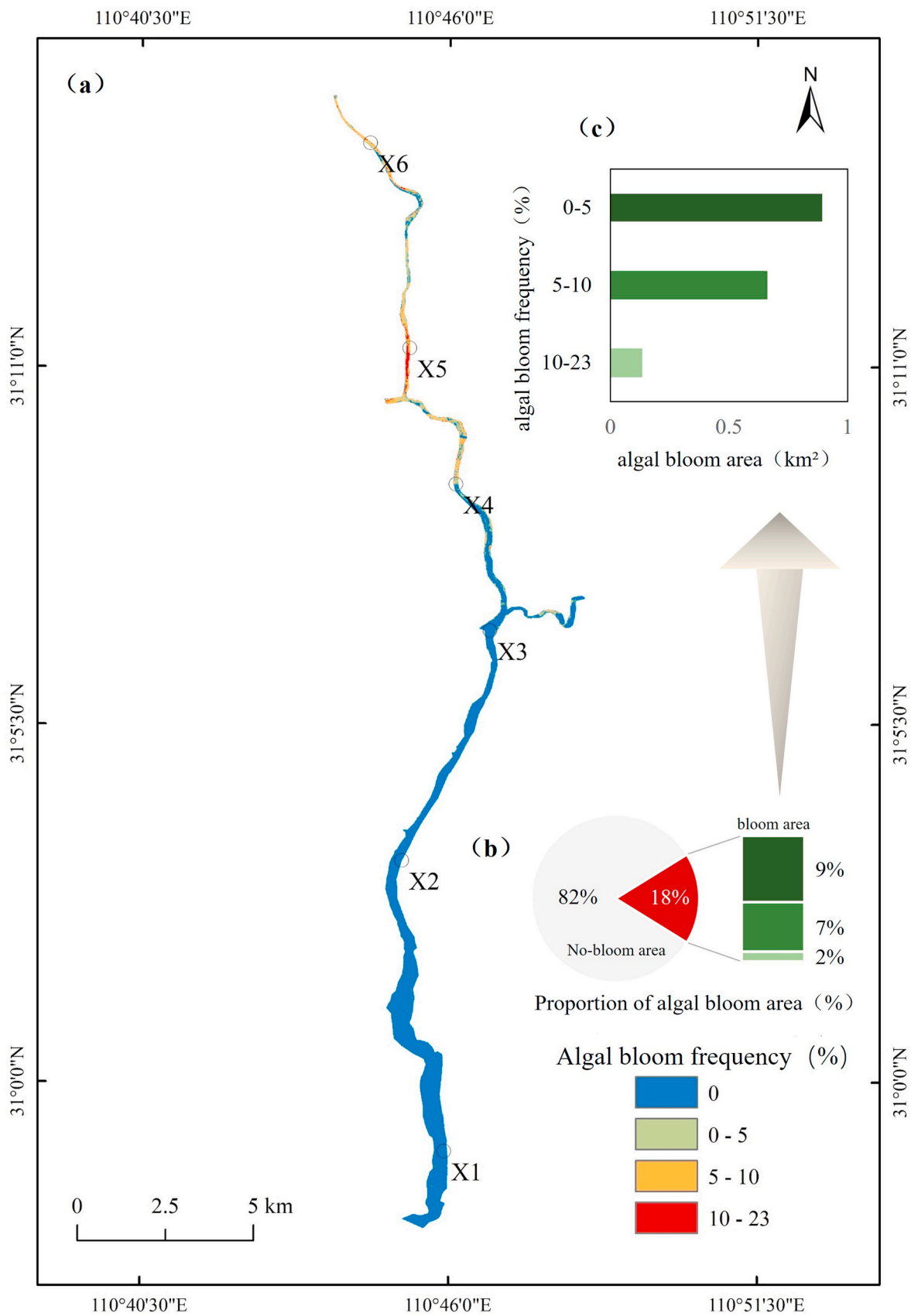


Fig. 8. Algal bloom frequency distribution and area proportion in XXB from 2019 to 2023.
(a) Algal bloom frequency distribution; (b) Proportion of algal bloom area; (c) Area associated with different algal bloom frequencies.

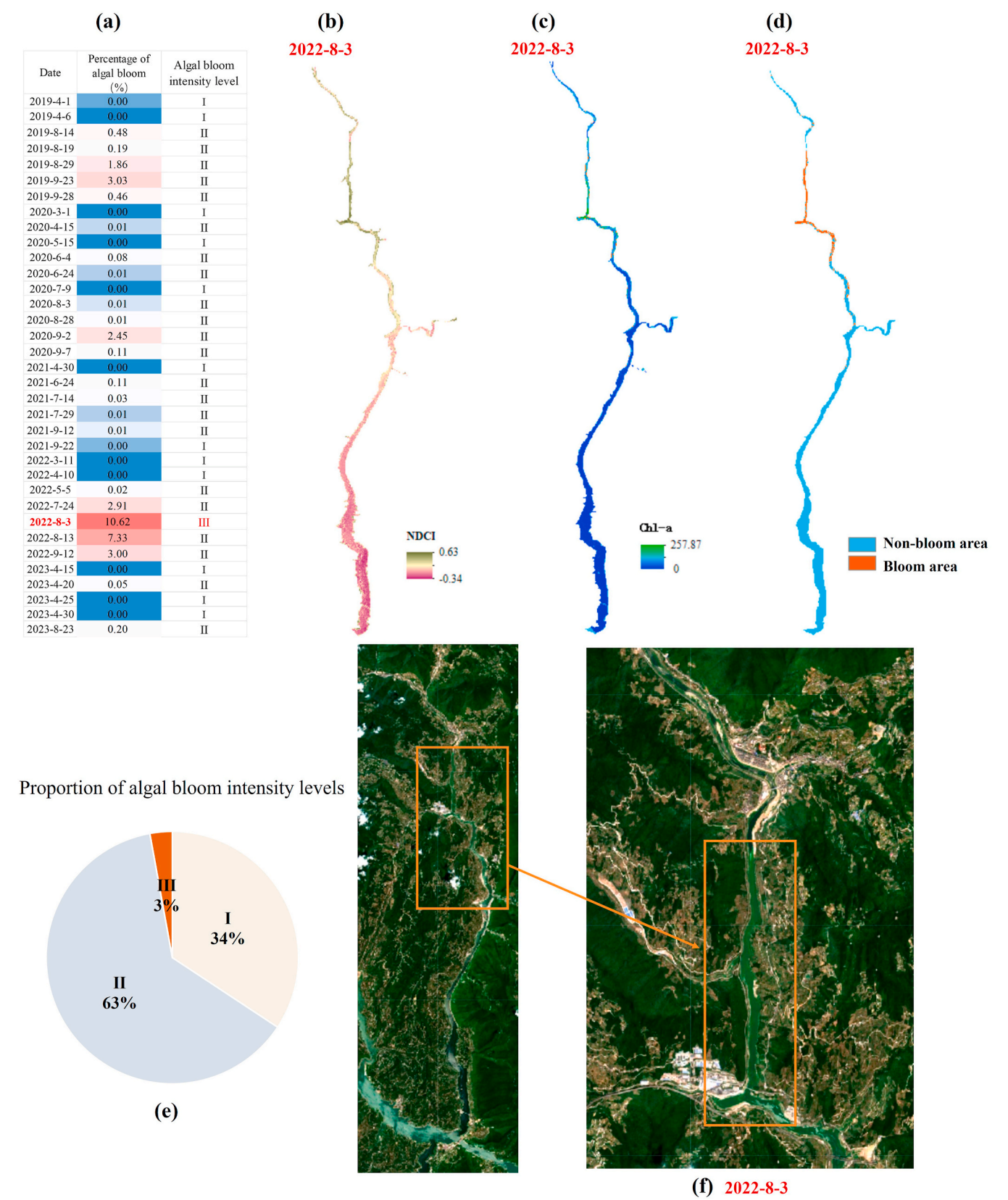


Fig. 9. Algal bloom intensity classification and remote sensing retrieval of typical algal bloom events in the XXB. (a) Algal bloom intensity level in 2019–2023; (b)–(d) Remote sensing retrieval of typical algal bloom events in the XXB on August 3, 2022; (e) Proportion of algal bloom intensity; (f) True-color remote sensing view of the mid-upper XXB during the bloom outbreak on August 3, 2022.

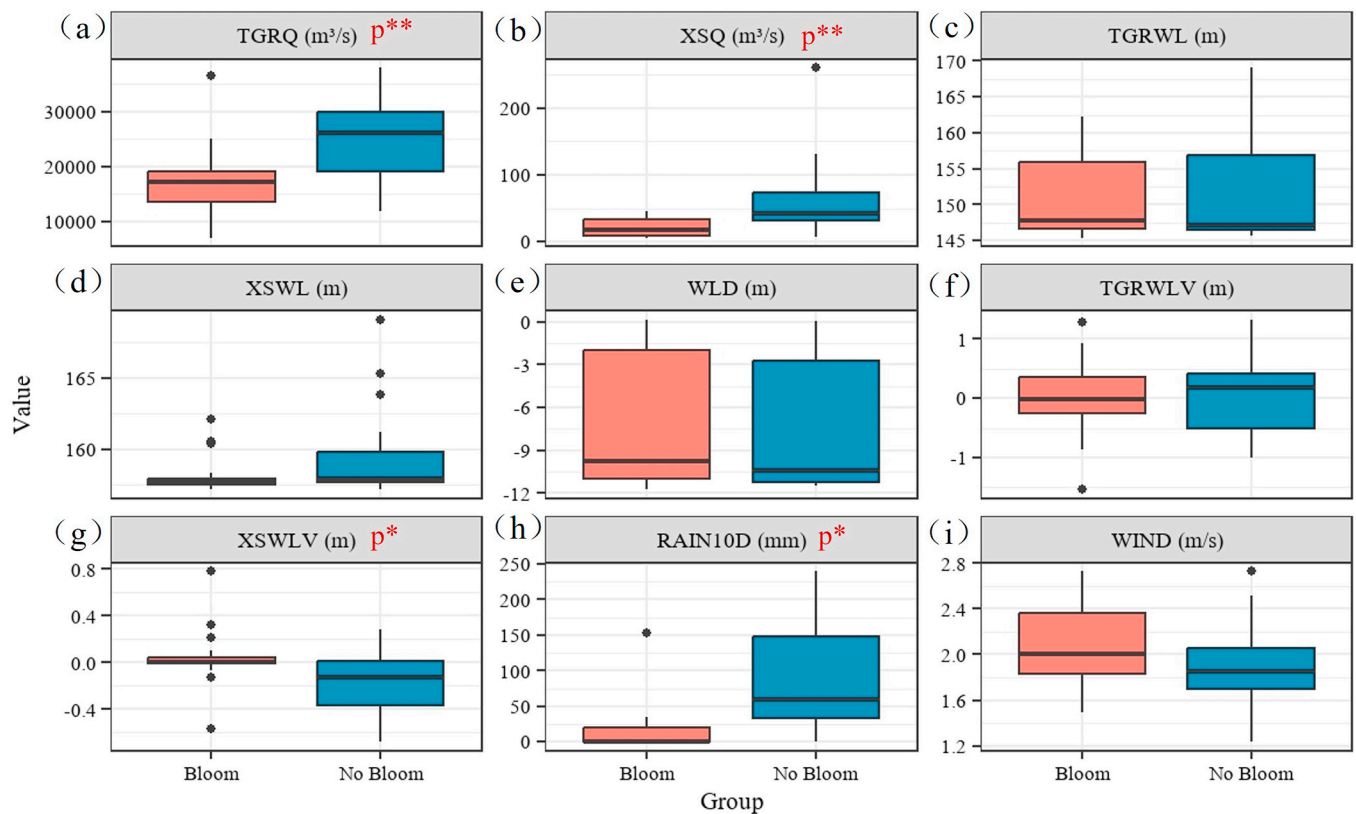


Fig. 10. Boxplots of hydrometeorological factors under bloom and non-bloom. Note: p indicates the significance level, *p < 0.05, **p < 0.01, ***p < 0.001.

identified as an effective spectral index for Chl-a retrieval in the XXB. This study further demonstrates that Sentinel-2 imagery is an effective tool for monitoring algal blooms in the narrow tributary bays of large reservoirs. When combined with in situ sampling data, the NDCI index yielded the most accurate prediction of Chl-a concentrations in the tributary, achieving a moderate to strong coefficient of determination ($R^2 = 0.76$).

Moreover, among Sentinel-2's multispectral bands, four visible bands (B2, B3, B4, and B8) offer a spatial resolution of 10 m. This allows for effective coverage of large water bodies such as lakes and reservoirs while enabling clear delineation of boundaries, patch distributions, and algal blooms in small tributaries. In addition to its high spatial resolution, Sentinel-2 also provides excellent temporal resolution. With twin satellites operating in tandem, the revisit interval is reduced to five days, facilitating frequent observations of the same area and allowing for timely detection of bloom onset, progression, and dissipation. This high revisit frequency makes Sentinel-2 particularly suitable for emergency monitoring and early warning of sudden bloom events. In comparison, Landsat has a 16-day revisit cycle, with reduced monitoring efficiency under cloudy conditions. Although MODIS offers daily observations, its coarse spatial resolution limits its utility for detailed bloom monitoring.

Beyond the approach adopted in this study—retrieving Chl-a by integrating Sentinel-2 spectral indices with in situ measurements—other algorithms for satellite-derived Chl-a estimation also warrant consideration. The Case 2 Regional Coast Colour (C2RCC) processor in ESA's SNAP platform employs a full-spectrum neural-network inversion trained on coupled atmosphere–water radiative-transfer simulations, enabling the retrieval of inherent optical properties and Chl-a concentrations in Case-II waters (Izurieta et al., 2023; Brockmann et al., 2016). Owing to its compatibility with Sentinel-2 MSI and its demonstrated effectiveness in turbid and eutrophic inland environments, C2RCC represents a valuable complementary technique (Soriano-González et al., 2022). However, C2RCC was not applied in the present study due to its

stringent requirements for radiometric calibration accuracy and atmospheric correction, while the complex water color composition and pronounced spatiotemporal heterogeneity of the study area may introduce additional uncertainties. Moreover, the lack of in situ optical property measurements limits the feasibility of adequate regional tuning and independent validation. Therefore, an empirical spectral index-based approach calibrated with in situ Chl-a data was adopted to ensure stable and comparable retrieval performance at the scale of the study area. Nevertheless, incorporating C2RCC into future studies and conducting comparative analyses with empirical methods could further validate or enhance the robustness of Chl-a retrievals for the XXB and other tributaries of the TGR.

4.2. Distinctive drivers of algal bloom spatiotemporal patterns in XXB

Due to density differences caused by water temperature, the interaction between backflow from the Yangtze River and inflow from the tributary exhibits a distinctive gravitational flow, leading to the formation of different hydrodynamic zones in the XXB (Fig. 11a). From the estuary to the upstream tributary, these zones are classified as the exchange zone, fluctuation zone, retention zone, transition zone and riverine zone. This unique hydrodynamic zonation governs the spatial distribution patterns of algal blooms in the XXB. Analysis in Section 3.3 indicated that algal blooms in XXB mainly occurred in the mid-to upper reaches of the bay, which are located in the backflow retention zone, and predominantly took place during the flood season.

During the flood season, the mainstem water of the Yangtze River intrudes into the tributary from the mid-layer, while the upstream tributary water flows out along the bottom under the influence of gravitational flow (Liu et al., 2016; Zeng et al., 2022). These two flows meet in the mid-to upper reaches, forming a circulation that blocks further mixing. As a result, the intruding mainstem water splits into two branches: one reverses and exits toward the Yangtze River at the surface,

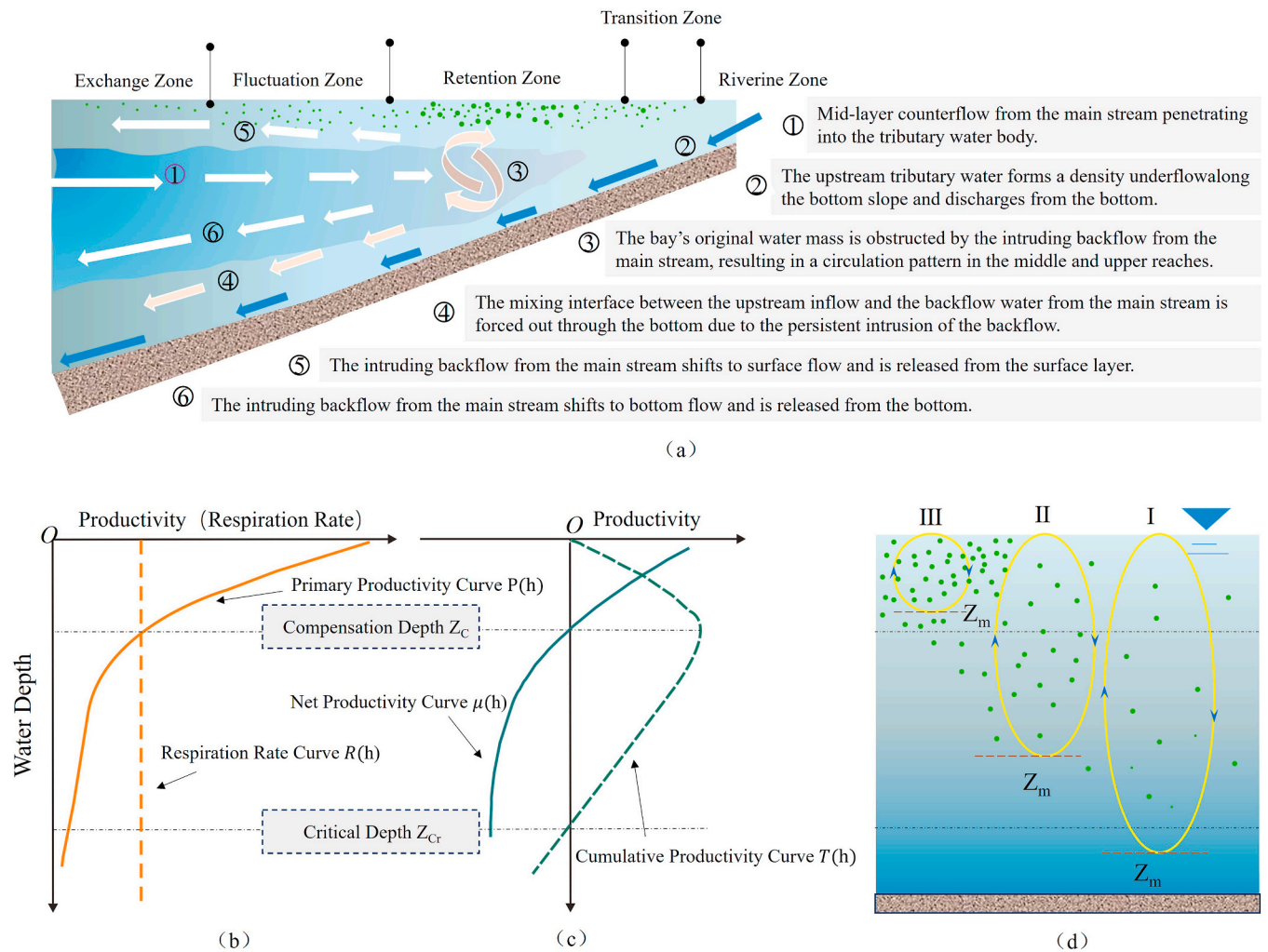


Fig. 11. Flood-season mixing patterns in XXB and the critical layer theory of algal bloom dynamics (Liu et al., 2016; Zeng et al., 2022) (a) Hydrodynamic mixing patterns of the XXB during the flood season; (b) Underwater respiration and photosynthesis; (c) Characterization of underwater stratification; (d) Relationship between mixed layer depth and algal blooms.

while the other, influenced by the circulation, exits at the mid-to lower layers. In addition, the interface between the mainstem and tributary waters flows out along the bottom under the continuous intrusion of the mainstem water (Fig. 11a). This stratified gravitational flow not only creates distinct layering in the shallow areas away from the estuary and gradually reduces the mixed layer depth (Z_m) along the bay toward the upstream, but also continuously delivers nutrients from the intruding mainstem water to the tributary, which are available for algal growth. Meanwhile, in the mid-to upper retention zone, flow obstruction leads to reduced velocities; the slow-moving water promotes rapid sedimentation, increasing water transparency and deepening the critical layer. At the depth corresponding to the critical layer (Z_{Cr}), the total phytoplankton production over 24 h equals the total loss. According to the critical layer theory, net biomass accumulation occurs only when Z_m is shallower than the Z_{Cr} (Sverdrup, 1953) (Fig. 11b and c).

Drawing on Sverdrup's critical depth theory for predicting spring phytoplankton blooms in the ocean (Sverdrup, 1953), a "critical layer theory" for phytoplankton growth and dissipation in XXB has been proposed. This theory has been extensively validated through field observations and modeling, and has become a classical framework for understanding algal blooms in the tributaries of the TGR (Liu et al., 2016; Yang, 2014; Yang et al., 2017b; Zeng et al., 2022). According to the critical layer theory, once the Z_m in the bay becomes shallower than the Z_{Cr} , phytoplankton can receive sufficient light for growth, leading to

bloom formation. During the flood season, stratified gravitational flow in the bay creates distinct layering in the mid-to upper reaches. When both the inflow to the TGR and the upstream tributary discharge are relatively low, the Z_m can easily become shallower than the Z_{Cr} , allowing phytoplankton to start proliferating, posing a risk of bloom development. Once the Z_m becomes shallower than the compensation depth (Z_C), phytoplankton are exposed to ample light, leading to rapid growth and bloom outbreaks (Fig. 11b,c,d). In this study, algal blooms primarily occurred during the flood season under low-flow conditions, predominantly in the mid-upper reaches of the tributary bays, which was attributed to the distinctive hydrodynamic-algal ecological mechanism.

4.3. Limitations of the study and implications

Based on remote sensing imagery from bloom periods between 2019 and 2023, this study employed the NDCI spectral index to retrieve Chl-a concentrations and identify bloom in XXB. At a broad spatial scale, the spatiotemporal dynamics of blooms in riverine bay areas were examined. Monthly standardized anomalies of Chl-a (Fig. 7) facilitated interpretation of seasonal variations and the influence of different reservoir operation periods, while bloom frequency distribution maps (Fig. 8) highlighted spatial patterns of blooms, offering insights for intensified monitoring and targeted management.

However, for inland water bodies, particularly narrow rivers,

obtaining high spatiotemporal resolution remote sensing imagery that coincides with sampling dates remains challenge. Weather conditions strongly affect the quality of the imagery, making it difficult to acquire images corresponding to the sampling periods (Gholizadeh et al., 2016). Therefore, based on the observed bloom distribution patterns, intensifying in situ monitoring during peak bloom periods, combined with extensive field measurements and spectral index data, can support the development of machine learning and deep learning models for bloom retrieval and prediction, further improving model accuracy. In addition, future technological applications could explore innovative approaches that integrate Sentinel-2 imagery with other multi-source remote sensing data to generate composite products with both high spatial and temporal resolution. This approach can fully integrate the unique advantages of multi-source data in terms of spatial detail, temporal continuity, and spectral information, ultimately producing high spatiotemporal resolution remote sensing imagery for the target area and addressing the limitations of coarse-resolution imagery in narrow river systems (Yang et al., 2022b).

In this study, the formation mechanism of algal blooms was explained from remote sensing imagery and combined with hydrometeorological data, the hydrodynamic mixing patterns of XXB, and the critical depth theory of bloom growth and dissipation in the TGR tributaries to analyze the external and internal drivers of bloom outbreaks in the mid-to upper reaches of XXB during flood seasons. The results indicate that the unique backflow intrusion pattern of the bay shapes the spatiotemporal distribution of blooms: the interaction between the mainstem and tributary flows in the mid-to upper reaches creates a stratified retention zone, where slow flow and prolonged low rainfall conditions (10-day cumulative rainfall < 35 mm) provide a stable hydrodynamic environment for algal growth. This stabilizes the Z_{Cr} for phytoplankton growth, and during the flood season, particularly under low reservoir inflow and reduced upstream tributary discharge, the Z_m becomes shallower. Once the Z_m is less than the Z_C , phytoplankton receive sufficient light to proliferate rapidly, triggering bloom outbreaks.

Therefore, Future management of algal blooms in XXB could adopt a combination of regional physical disturbance and basin-scale water management strategies. First, regional physical disturbance measures, such as artificial mixing, can be applied to disrupt the stable flow conditions in the mid-to upper reaches of the XXB. Second, under the operational constraints of flood control, hydropower generation, navigation, and water supply in the TGR, ecological regulation in coordination with upstream cascade reservoirs can be implemented. During flood-season periods when the 10-day cumulative rainfall is below 35 mm, reservoir inflow can be increased and water levels adjusted to increase the depth of Z_m and enhance water column disturbance, thereby breaking conditions favorable for algal proliferation and suppressing bloom development.

5. Conclusion

This study developed an efficient approach for monitoring algal bloom dynamics in narrow riverine bays of the TGR using long-term Sentinel-2 imagery (2019–2023), with XXB as a representative case. The underlying hydrometeorological mechanisms and corresponding mitigation strategies were also clarified. The main conclusions are as follows.

- (1) Sentinel-2 imagery enables reliable, fine-scale monitoring of algal blooms in tributary backwater zones. The NDCI index shows strong performance in Chl-a retrievals in XXB ($R^2 = 0.76$), and bloom events can be effectively detected using Chl-a thresholds. High-density in situ Chl-a measurements remain essential for sustaining retrieval accuracy.
- (2) Algal blooms in XXB primarily occur in March, July–September, with August being the peak month. About 18 % of the bay has

experienced blooms, mainly in the mid-to upper backwater reaches, with the region near X5 (Pingyikou) showing the highest intensity and frequency. Bloom intensity is dominated by levels II and III.

- (3) During the flood season, mid-layer intrusion from the Yangtze River and upstream tributary inflow jointly create stable stratification in XXB, especially under low-rainfall conditions. Reduced mainstem and tributary discharge further limits vertical mixing and favors bloom formation.
- (4) Bloom mitigation should focus on the mid-to upper reaches of the bay. Physical disturbance (e.g., artificial mixing) and ecological regulation. Under low 10-day rainfall (< 35 mm), reservoir inflow and water-level operations can enhance mixing and suppress bloom development.

CRediT authorship contribution statement

Aiping Jiang: Writing – original draft, Visualization, Software, Methodology. **Dongsheng Wang:** Supervision, Conceptualization. **Tiantian Jin:** Writing – review & editing, Methodology. **Peng Li:** Funding acquisition. **Tao Xu:** Funding acquisition. **Di Zhang:** Writing – review & editing. **Boran Zhu:** Visualization, Software. **Junqiang Lin:** Methodology, Formal analysis. **Qidong Peng:** Supervision, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgment

This work was supported by funding from China Yangtze Power Co., Ltd. scientific research project (Z242302042).

Data availability

The data that has been used is confidential.

References

- Aláez, Bellas, Francisco, Torres, Jesús, Spyarakos, Evangelos, Vilas, Luis Gonzalez, 2021. Machine learning methods applied to the prediction of Pseudo-nitzschia spp. blooms in the Galician rias baixas (NW Spain). *ISPRS Int. J. Geoinf.* 10, 199. <https://doi.org/10.3390/ijgi10040199>.
- Ashour, Mohammed, Guan, Di, Zheng, Guochen, 2021. Influencing factors analysis for algal blooms in cold region reservoir. *J. Phys., Conf. Ser.* 1732, 012147. <https://doi.org/10.1088/1742-6596/1732/1/012147>.
- Beal, Max, Ozdogan, Mutlu, Paul, Block, 2024. A machine learning and remote sensing-based model for algae pigment and dissolved oxygen retrieval on a small inland Lake. *Water Resour. Res.* 60. <https://doi.org/10.1029/2023WR035744>.
- Brockmann, Carsten, Doerffer, Roland, Peters, Marco, Stelzer, Kerstin, Embacher, Sabine, Ruescas, Ana, 2016. Evolution of the C2RCC neural network for Sentinel 2 and 3 for the retrieval of ocean colour products in normal and extreme optically complex waters. *Environ. Sci., Comput. Sci., Eng.*
- Caballero, Isabel, Gabriel, Navarro, 2021. Monitoring cyanoHABs and water quality in Laguna Lake (Philippines) with Sentinel-2 satellites during the 2020 Pacific typhoon season. *Sci. Total Environ.* 788, 147700. <https://doi.org/10.1016/j.scitotenv.2021.147700>.
- Chuo, Mingying, Ma, Jun, Liu, Defu, Yang, Zhengjian, 2019. Effects of the impounding process during the flood season on algal blooms in Xiangxi Bay in the Three Gorges Reservoir, China. *Ecol. Model.* 392, 236–249. <https://doi.org/10.1016/j.ecolmodel.2018.11.017>.
- Elhag, Mohamed, Gitas, Ioannis, Othman, Anas, Bahrawi, Jarbou, Gikas, Petros, 2019. Assessment of water quality parameters using temporal remote sensing spectral reflectance in arid environments. *Saudi Arabia. Water.*
- Gao, Qifeng, He, Guojian, Fang, Hongwei, Bai, Sen, Huang, Lei, 2018. Numerical simulation of water age and its potential effects on the water quality in Xiangxi Bay of Three Gorges Reservoir. *J. Hydrol.* 566, 484–499. <https://doi.org/10.1016/j.jhydrol.2018.09.033>.

- Gholizadeh, Mohammad H., Melesse, Assefa M., Reddi, Lakshmi, 2016. A comprehensive review on water quality parameters estimation using remote sensing techniques. *Sensors* 16 (8), 1298. <https://doi.org/10.3390/s16081298>.
- Giri, Subhasis, 2021. Water quality perspective in Twenty First century: status of water quality in major river basins, contemporary strategies and impediments: a review. *Environ. Pollut.* 271, 116332. <https://doi.org/10.1016/j.envpol.2020.116332>.
- Gorelick, Noel, Hancher, Matt, Dixon, Mike, Ilyushchenko, Simon, Thau, David, Moore, Rebecca, 2017. Google Earth engine: planetary-scale geospatial analysis for everyone. *Rem. Sens. Environ.* 202, 18–27. <https://doi.org/10.1016/j.rse.2017.06.031>.
- Gower, Jim, King, S., Goncalves, P., 2008. Global monitoring of plankton blooms using MERIS MCI. *Int. J. Rem. Sens.* 29, 6209–6216. <https://doi.org/10.1080/01431160802178110>.
- Hossen, Hickmat, Mahmood, Wael Elham, Negm, Abdelazim, Nakamura, Takashi, 2022. Assessing water quality parameters in burullus Lake using Sentinel-2 satellite images. *Water Resour.* 49, 321–331. <https://doi.org/10.1134/S0097807822020087>.
- Huo, Shouliang, Zhang, Hanxiao, Monchamp, Marie-Eve, Wang, Rong, Weng, Nanyan, Zhang, Jingting, Zhang, Hong, Wu, Fengchang, 2022. Century-long homogenization of algal communities is accelerated by nutrient enrichment and climate warming in Lakes and reservoirs of the north temperate zone. *Environ. Sci. Technol.* 56, 3780–3790. <https://doi.org/10.1021/acs.est.1c06958>.
- Institute of Hydrobiology, Chinese Academy of Sciences, 2024. Analysis Report on the Monitoring Results of Algal Blooms in the Three Gorges Reservoir from 2021 to 2023 (in Chinese).
- Izurrieta, Ayala, Johanna, Dávalos, Andrés, Jara, Carlos, Ponce, Sofia Godoy, Van Wittenberghe, Shari, Verrelst, Jochem, Delegido, Jesús, 2023. Spatial and temporal analysis of water quality in high andean Lakes with Sentinel-2 satellite automatic water products. *Sensors* 23, 8774. <https://doi.org/10.3390/s23218774>.
- Jiang, Aiping, Wang, Dongsheng, Ning, Zhen, Peng, Li, Xu, Tao, Zhu, Boran, Peng, Qidong, Jin, Tiantian, Lin, Junqiang, Zhang, Di, 2025. Spatio-temporal characteristics and influencing factors of water quality in Xiangxi Bay under the operation of the three gorges reservoir. *J. Contam. Hydrol.* 270, 104518. <https://doi.org/10.1016/j.jconhyd.2025.104518>.
- Jin, Xiangang, Scott, Wells, Liu, De, Yang, Guolu, 2018. Thermal stratification and its relationship with water quality in the typical tributary bay of the Three Gorges Reservoir. *Water Sci. Technol. Water Supply* 19. <https://doi.org/10.2166/ws.2018.142>.
- Khalilji, Mohammad Hossein, Hasanlou, Mahdi, 2019. Harmful algal blooms monitoring using Sentinel-2 satellite images. *ISPRS - Int. Arch. Photogram., Rem. Sens. Spat. Inform. Sci.* 609–613. <https://doi.org/10.5194/isprs-archives-XLII-4-W18-609-2019>. XLII-4/W18.
- Kim, Woo, Young, TaeHo Kim, Shin, Jihoon, Lee, Dae-Seong, Park, Young-Seuk, Kim, Yeji, Cha, YoonKyung, 2022. Validity evaluation of a machine-learning model for chlorophyll a retrieval using Sentinel-2 from inland and coastal waters. *Ecol. Indic.* 137, 108737. <https://doi.org/10.1016/j.ecolind.2022.108737>.
- Kislik, Chippie, Dronova, Iryna, Grantham, Theodore E., Kelly, Maggi, 2022. Mapping algal bloom dynamics in small reservoirs using Sentinel-2 imagery in Google Earth Engine. *Ecol. Indic.* 140, 109041. <https://doi.org/10.1016/j.ecolind.2022.109041>.
- Kwon, Yong Sung, Pyo, JongCheol, Kwon, Yong-Hwan, Duan, Hongtao, Cho, Kyung Hwa, Park, Yongeun, 2020. Drone-based hyperspectral remote sensing of Cyanobacteria using vertical cumulative pigment concentration in a deep reservoir. *Rem. Sens. Environ.* 236, 111517. <https://doi.org/10.1016/j.rse.2019.111517>.
- Li, Peiyao, Yao, Ye, Lian, Jijian, Ma, Chao, 2021a. Effect of thermal stratified flow on algal blooms in a tributary bay of the Three Gorges reservoir. *J. Hydrol.* 601, 126648. <https://doi.org/10.1016/j.jhydrol.2021.126648>.
- Li, Sijia, Song, Kaishan, Wang, Shuai, Liu, Ge, Wen, Zhidan, Shang, Yingxin, Lyu, Lili, Chen, Fangfang, Xu, Shiqi, Tao, Hui, Du, Yunxia, Fang, Chong, Mu, Guangyi, 2021b. Quantification of chlorophyll-a in typical lakes across China using Sentinel-2 MSI imagery with machine learning algorithm. *Sci. Total Environ.* 778, 146271. <https://doi.org/10.1016/j.scitotenv.2021.146271>.
- Li, Yiping, Huang, Yanan, Ji, Daobin, Cheng, Yu, Nwankwegu, Amechi S., Paerl, Hans W., Tang, Chunyan, Yang, Zhengjian, Zhao, Xingxing, Chen, Yi, Li, Jinhua, 2022. Storm and floods increase the duration and extent of phosphorus limitation on algal blooms in a tributary of the Three Gorges Reservoir, China. *J. Hydrol.* 607, 127562. <https://doi.org/10.1016/j.jhydrol.2022.127562>.
- Liaw, Andy, Wiener, Matthew C., 2007. Classification and regression by randomForest. In: *Classification and Regression by Random Forest*.
- Lichtenthaler, Hartmut, 1987. Chlorophylls and carotenoids: pigments of photosynthetic biomembranes. *Methods Enzymol.* 148C, 350–382. [https://doi.org/10.1016/0076-6879\(87\)48036-1](https://doi.org/10.1016/0076-6879(87)48036-1).
- Liu, Defu, Yang, Zhengjian, Ji, Daobin, Ma, Jun, Cui, Yujie, Song, Linxu, 2016. A review on the mechanism and its controlling methods of the algal blooms in the tributaries of Three Gorges Reservoir. *J. Hydraul. Eng.* 47, 443–454 (in Chinese).
- Liu, Haiqiu, Ren, Hengkui, Niu, Xinxin, Xia, Ping, 2021. Extraction of Cyanobacteria bloom in Chaohu Lake based on Sentinel-2 remote sensing images. *Ecol. Environ. Sci.* 30, 146–155 (in Chinese).
- Lobo, Felipe D., Nagel, Gustavo W., Maciel, Daniel A., Carvalho, Lino A., Martins, Vitor S., Barbosa, Cláudio C., Novo, Evelyn M., 2021. AlgaeMAP: algae bloom monitoring application for Inland waters in Latin America. *Remote Sens.* 13 (15), 2874. <https://doi.org/10.3390/rs13152874>.
- Luo, Wenguang, Luo, Xin, Lu, Jing, Ming, Bo, 2022. Contribution of the reservoir backflow to the eutrophication of its tributary: a case study of the Xiangxi River, China. *Hydrol. Res.* 53. <https://doi.org/10.2166/nh.2022.122>.
- Ma, Jun, Liu, Defu, Wells, Scott A., Tang, Hongwu, Ji, Daobin, Yang, Zhengjian, 2015. Modeling density currents in a typical tributary of the Three Gorges Reservoir, China. *Ecol. Model.* 296, 113–125. <https://doi.org/10.1016/j.ecolmodel.2014.10.030>.
- Ma, Jieying, Jin, Shuanggen, Li, Jian, He, Yang, Shang, Wei, 2021. Spatio-Temporal variations and driving forces of harmful algal blooms in Chaohu Lake: a multi-source remote sensing approach. *Remote Sens.* 13 (3), 427. <https://doi.org/10.3390/rs13030427>.
- Ministry of Ecology and Environment of the People's Republic of China, 2017. *Water Quality—Determination of Chlorophyll a—Spectrophotometric Method (HJ 897-2017)* (in Chinese).
- Ministry of Ecology and Environment of the People's Republic of China, 2020. *Technical Specifications for Monitoring and Evaluating Algal Bloom Based on Remote Sensing and Field Monitoring* (in Chinese).
- Mishra, Sachidananda, Mishra, Deepak R., 2012. Normalized difference chlorophyll index: a novel model for remote estimation of chlorophyll-a concentration in turbid productive waters. *Rem. Sens. Environ.* 117, 394–406. <https://doi.org/10.1016/j.rse.2011.10.016>.
- Mulligan, Mark, van Soesbergen, Arnout, Sáenz, Leonardo, 2020. GOODD, a global dataset of more than 38,000 georeferenced dams. *Sci. Data* 7, 31. <https://doi.org/10.1038/s41597-020-0362-5>.
- Peng, Kaida, Jiao, Yiyi, Gao, Jian, Wen, Xiong, Zhao, Yijun, Yang, Shao, Liao, Mingjun, 2023. Viruses may facilitate the cyanobacterial blooming during summer bloom succession in Xiangxi Bay of Three Gorges Reservoir, China. *Front. Microbiol.* 14, 1112590. <https://doi.org/10.3389/fmicb.2023.1112590>.
- Qing, Song, Jin, Eerdemutu, Hao, Yanling, Zhao, Wenjing, 2021. A spectral index for the detection of algal blooms using Sentinel-2 Multispectral Instrument (MSI) imagery: a case study of Hulun Lake, China. *Int. J. Rem. Sens.* 42, 4514–4535. <https://doi.org/10.1080/01431161.2021.1897186>.
- Reid, Andrea J., Carlson, Andrew K., Creed, Irena F., Eliason, Erika J., Gell, Peter A., Johnson, Pieter T.J., Kidd, Karen A., MacCormack, Tyson J., Olden, Julian D., Ormerod, Steve J., Smol, John P., Taylor, William W., Tockner, Klement, Vermaire, Jesse C., Dudgeon, David, Cooke, Steven J., 2019. Emerging threats and persistent conservation challenges for freshwater biodiversity. *Biol. Rev.* 94, 849–873. <https://doi.org/10.1111/brev.12480>.
- Salman, Hasan, Ali, Kalakech, Steiti, Amani, 2024. Random forest algorithm overview. *Babylonian J. Machine Learn.* 69–79. <https://doi.org/10.58496/BJML/2024/007>.
- Sang, Chong, Tan, Lu, Cai, Qinghua, Ye, Lin, 2024. Long-term (2003–2021) evolution trend of water quality in the Three Gorges Reservoir: an evaluation based on an enhanced water quality index. *Sci. Total Environ.* 915, 169819. <https://doi.org/10.1016/j.scitotenv.2023.169819>.
- Shi, Kun, Zhang, Yunlin, Zhou, Yongqiang, Liu, Xiaohan, Zhu, Guangwei, Qin, Boqiang, Gao, Guang, 2017. Long-term MODIS observations of cyanobacterial dynamics in Lake Taihu: responses to nutrient enrichment and meteorological factors. *Sci. Rep.* 7, 40326. <https://doi.org/10.1038/srep40326>.
- Shi, Xuming, Gu, Lingjia, Jiang, Tao, Zheng, Xingming, Dong, Wen, Tao, Zui, 2022. Retrieval of Chlorophyll-a concentrations using Sentinel-2 MSI imagery in Lake Chagan based on assessments with machine learning models. *Remote Sens.* 14 (19), 4924. <https://doi.org/10.3390/rs14194924>.
- Soriano-González, Jesús, Urrego, Esther P., Soria-Perpinya, Xavier, Angelats, Eduard, Alcaraz, Carles, Delegido, Jesús, Ruiz-Verdú, Antonio, Tenjo, Carolina, Vicente, Eduardo, Moreno, José, 2022. Towards the combination of C2RCC processors for improving water quality retrieval in inland and coastal areas. *Remote Sens.* 14 (5), 1124. <https://doi.org/10.3390/rs14051124>.
- Sverdrup, H.U., 1953. On conditions for the vernal blooming of phytoplankton. *Journal du Conseil.* 18, 287–295. <https://doi.org/10.1093/icesjms/18.3.287>.
- Toming, Kaire, Kutser, Tiit, Laas, Alo, Sepp, Margot, Paavel, Birgot, Noges, Tiina, 2016. First experiences in mapping LakeWater quality parameters with Sentinel-2 MSI imagery. *Rem. Season.* 8. <https://doi.org/10.3390/rs8080640>.
- Wang, Xiaoyan, Yang, Wu, 2019. Water quality monitoring and evaluation using remote-sensing techniques in China: a systematic review. *Ecosys. Health Sustain.* 5. <https://doi.org/10.1080/20964129.2019.1571443>.
- Wang, Xiaodi, Song, Kaishan, Wen, Zhidan, Liu, Ge, Shang, Yingxin, Fang, Chong, Lyu, Lili, Wang, Qiang, 2021. Quantifying turbidity variation for Lakes in daqing of northeast China using landsat images from 1984 to 2018. *IEEE J. Sel. Top. Appl. Earth Obs. Rem. Sens.* 14, 8884–8897. <https://doi.org/10.1109/JSTARS.2021.3101475>.
- Wang, Dong, Tang, Bo-Hui, Huang, Liang, Li, Menghua, Chen, Guokun, Pan, Xuejun, 2022. Estimation of Chlorophyll-A concentration with remotely sensed data for the nine Plateau Lakes in Yunnan province. *Remote Sens.* 14, 4950. <https://doi.org/10.3390/rs14194950>.
- Wang, Mengru, Bodirsky, Benjamin Leon, Rinjevel, Rhodé, Beier, Felicitas, Bak, Mirjam P., Batool, Masooma, Droppers, Bram, Popp, Alexander, van Vliet, Michelle T.H., Strokal, Maryna, 2024. A triple increase in global river basins with water scarcity due to future pollution. *Nat. Commun.* 15, 880. <https://doi.org/10.1038/s41467-024-44947-3>.
- Wang, Wanting, Wang, Guoqiang, Li, Jie, Chen, Jinyue, Gao, Zhenyu, Fang, Lei, Ren, Shilong, Wang, Qiao, 2025. Remote sensing identification and model-based prediction of harmful algal blooms in inland waters: current insights and future perspectives. *Water Res.* X 28, 100369. <https://doi.org/10.1016/j.wroa.2025.100369>.
- Weber, Samuel J., Mishra, Deepak R., Wilde, Susan B., Kramer, Elizabeth, 2020. Risks for cyanobacterial harmful algal blooms due to land management and climate interactions. *Sci. Total Environ.* 703, 134608. <https://doi.org/10.1016/j.scitotenv.2019.134608>.
- Xiao, Xi, Agustí, Susana, Pan, Yaoru, Yu, Yan, Li, Ke, Wu, Jiaping, Duarte, Carlos M., 2019. Warming amplifies the frequency of harmful algal blooms with eutrophication

- in Chinese coastal waters. *Environ. Sci. Technol.* 53, 13031–13041. <https://doi.org/10.1021/acs.est.9b03726>.
- Yang, Zhengjian, 2014. The Mechanisms of Algal Blooms and its Operation Method Through Water Level Fluctuation Under the Situation of the Bidirectional Density Currents in Tributaries of the Three Gorges Reservoir. Wuhan University (in Chinese).
- Yang, Chaowei, Yu, Manzhu, Hu, Fei, Jiang, Yongyao, Li, Yun, 2017a. Utilizing cloud computing to address big geospatial data challenges. *Comput. Environ. Urban Syst.* 61, 120–128. <https://doi.org/10.1016/j.compenvurbsys.2016.10.010>.
- Yang, Zhengjian, Yu, Yan, Chen, Zhao, Jun, Ma, 2017b. Mechanism of eutrophication and phytoplankton blooms in Three Gorges Reservoir, China: a research review. *Eng. J. Wuhan Univ.* 50, 507–516 (in Chinese).
- Yang, Zhiming, Reiter, Michael, Munyei, Nicolas, 2017c. Estimation of chlorophyll-a concentrations in diverse water bodies using ratio-based NIR/red indices. *Remote Sens. Appl.: Soc. Environ.* 6, 52–58. <https://doi.org/10.1016/j.rsase.2017.04.004>.
- Yang, Haibo, Du, Yao, Zhao, Hongling, Chen, Fei, 2022a. Water quality Chl-a inversion based on spatio-temporal fusion and convolutional neural network. *Remote Sens.* 14 (5), 1267. <https://doi.org/10.3390/rs14051267>.
- Yang, Haibo, Kong, Jialin, Hu, Huihui, Du, Yao, Gao, Meiyan, Chen, Fei, 2022b. A review of remote sensing for water quality retrieval: progress and challenges. *Remote Sens.* 14, 1770. <https://doi.org/10.3390/rs14081770>.
- Yao, Jinzhong, Fan, Xiangjun, Yang, Xia, Huang, Yubo, 2022. Current situation, causes and control measures of water bloom in the key tributaries of the Three Gorges Reservoir. *Chin. J. Environ. Eng.* 16, 2041–2048 (in Chinese).
- Zeng, Yiheng, Shen, Xuzhou, Zhang, Jialei, Zhang, Huan, Liu, Defu, Gong, Chuan, weng, Chuansong, 2022. Effects of stratified density flow on the spatial distribution of chlorophyll-a in phytoplankton in Xiangxi River. *J. Environ. Eng. Technol.* 12, 426–435 (in Chinese).
- Zhang, Jiawei, Meng, Fei, Fu, Pingjie, Jing, Tingting, Xu, Jie, Yang, Xinyue, 2024. Tracking changes in chlorophyll-a concentration and turbidity in Nansi Lake using Sentinel-2 imagery: a novel machine learning approach. *Ecol. Inform.* 81, 102597. <https://doi.org/10.1016/j.ecoinf.2024.102597>.
- Zheng, Binghui, Zhang, Yuan, Fu, Guo, Liu, Hongliang, 2006. On the assessment standards for nutrition status in the Three Gorge Reservoir. *Acta Sci. Circumstantiae* 26, 1022–1030 (in Chinese).
- Zheng, Lilin, Xu, Jinying, Wang, Dandan, Xu, Guoyu, Tan, Zhiqiang, Xu, Ligang, Wang, Xiaolong, 2020. Acceleration of vegetation dynamics in hydrologically connected wetlands caused by Dam operation. *Hydrol. Process.* 35. <https://doi.org/10.1002/hyp.14026>.
- Zhi, Xiaosha, Chen, Lei, Chen, Shibo, Yu, Jiaqi, Jiang, Jing, Xu, Yanzhe, Li, Leifang, Meng, Xinyi, Shen, Zhenyao, 2024a. Heterogeneity and influencing factors of algal blooms in the reservoir-impacted tributary: evidence from remote sensing and physical-based model. *J. Hydrol.* 634, 131058. <https://doi.org/10.1016/j.jhydrol.2024.131058>.
- Zhi, Xiaosha, Shen, Zhenyao, Chen, Lei, Chen, Shibo, 2024b. The causes of algal blooms exist significant scale effect in tributary of the Three Gorges Reservoir. *J. Hydrol.* 640, 131677. <https://doi.org/10.1016/j.jhydrol.2024.131677>.
- Zhou, Botian, Shi, Kun, Wang, Weijia, Zhang, Dong, Qin, Boqiang, Zhang, Yunlin, Dong, Baili, Shang, Mingsheng, 2022. Phytoplankton succession phenology trends in the backwaters of the three gorges reservoir in China and their drivers: results from satellite observations. *Ecol. Indic.* 143, 109435. <https://doi.org/10.1016/j.ecolind.2022.109435>.